

Do Environmental Provisions in Trade Agreements Reshape Export Composition?

Evidence from the Universe of Chinese Customs Transactions, 2000–2015

Edoardo Vitella

University of Trento & Free University of Bozen

September 2026 — **preliminary draft**

Abstract

Environmental Provisions (EPs) have been increasingly embedded in Preferential Trade Agreements (PTAs) over the last two decades, with the expectation that they will play an even more important role in future agreements. However, empirical evidence on their trade effects relies almost entirely on aggregate bilateral flows. This paper asks whether EPs can shift export composition towards green goods and away from dirty goods, using a firm-product-destination-year panel for the universe of Chinese customs transactions spanning from 2000 to 2015. A triple-difference design compares green and dirty products against neutral counterparts within each firm-destination-year combination, with firm-destination-year fixed effects accounting for everything constant within each firm-market-year triple. The results indicate that there is no effect on the green margin, while the dirty margin is less clear-cut. The green null is robust to wild cluster bootstrap and permutation inference, and to replacing the comparison group. The dirty coefficient is negative almost everywhere but its statistical standing depends on which control for general agreement depth is used, and it does not survive a permutation test that reassigns the environmental content of the agreements at random across the treated destinations. What emerges is that the content of environmental provisions matters more than their mere presence. The PTAs signed by China over 2000–2015 contained mostly cooperation language, which makes our results consistent with the existing literature, where significant effects are driven by specific and enforceable clauses rather than loose language.

1 Introduction

Over the last decades, the number of preferential trade agreements (PTAs) has increased substantially, and many now include environmental provisions (EPs) or other forms of environmental language (Mattoo, Rocha, and Ruta, 2020; Morin, Pauwelyn, and Hollway, 2017). Because climate change is likely to become an increasingly important political concern, and given the evidence that global trade may play a role in environmental decay (Zhang et al., 2017), environmental content in trade agreements is likely to expand, since trade policy is one of the few levers available to address cross-border environmental externalities (Frankel, 2009). Accordingly, a growing literature studies the relationship between trade and the environment (Copeland, Shapiro, and Taylor, 2022), yet whether environmental provisions in trade agreements have a genuine effect on trade flows is still an open empirical question (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2025). This paper contributes to this literature by asking whether the environmental provisions included in preferential trade agreements signed by China between 2000 and 2015 shifted the composition of its exports toward greener products and away from dirtier ones, or whether those provisions left trade flows largely unaffected.

Most of the existing literature on the trade effects of environmental provisions in trade agreements comes from studies relying on aggregate bilateral trade flows, which makes it difficult to understand whether EPs cause a compositional shift in traded goods. In the first study of the trade effects of environmental provisions in PTAs, Brandi et al. (2020) find that EPs reduce dirty exports and increase green exports when regulations are specific and enforceable, especially for developing countries. On the other hand, Berger et al. (2020) use a gravity panel to show that membership in PTAs with more environmental provisions is associated with lower bilateral exports overall, with the effect being concentrated in South-North trade flows and not robust to PPML. Specific and enforceable clauses are also those that drive the effect in Abman, Lundberg, and Ruta, 2024, who find that forest preservation and biodiversity clauses in PTAs reduce deforestation in partner countries relative to those that do not include such clauses. Taken together, the evidence suggests that it is the content of the specific provisions and chapters that matters, rather than their mere presence in trade agreements. This paper brings the question to the largest exporter in the world, asking whether the environmental provisions in China’s PTAs between 2000 and 2015 had any effect on the composition of its exports, and whether the content of those provisions matters for trade flows.

Two identification problems are immediate. First, countries do not sign trade agreements at random, and the signing of agreements is correlated with pre-existing bilateral relation-

ships. Therefore, any comparison of export flows to PTA partners versus non-partners risks confounding the effect of environmental provisions with pre-existing differences in trade relationships. Second, aggregate bilateral trade flows cannot distinguish a reallocation of export composition toward greener goods from an expansion or contraction of trade that happens to coincide with agreement entry into force. To address the empirical question, this study uses the universe of Chinese customs transactions between 2000 and 2015, a 45-million-observation firm-product-destination-year panel with granularity at the HS6 product level, which makes it possible to track the same firm exporting the same product to multiple destinations simultaneously, and to observe how those flows diverge when one destination is covered by a preferential agreement and another is not.

Over the study period, China signed 14 PTAs with 25 destination economies, with environmental content ranging from a handful of generic cooperation clauses to dedicated chapters. Data on the content of the environmental provisions come from two independent sources, the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and the TREND database (Morin, Pauwelyn, and Hollway, 2017). The former provides a count of environmental-law provisions in force with each destination at each point in time, while the latter provides a more detailed coding of the content of those provisions, including sub-indices covering areas such as green market access and enforcement stringency. Using two independent codings guards against idiosyncrasies of either scheme and makes it possible to check the robustness of the results. We identify green goods on the basis of the OECD Combined List of Environmental Goods (CLEG) (Sauvage, 2014), while dirty goods are identified through a standard pollution-intensity classification that maps polluting industries to HS6 product codes, as developed in Mani and Wheeler, 1998. Everything else serves as the neutral comparison group.

The empirical design compares how exports of green and dirty products react to environmental provisions, relative to neutral products, within the same firm, destination and year. A triple-difference with a rich set of fixed effects, including firm-destination-year fixed effects, absorbs everything that is constant within a firm’s relationship with a specific destination market in a given year, including the agreement itself and any aggregate demand or supply shocks at the bilateral level. Because EPs enter into force with the agreement itself and rarely change their depth or content afterwards, EP depth and PTA depth are highly correlated, as agreements with more EP content tend to be also deeper agreements overall. Therefore, all specifications control for PTA depth separately, so that the coefficient on environmental provisions reflects environmental content proper. In this design, identification comes from the differential sensitivity of green and dirty products to environmental provisions, relative to neutral products, which serve as a within-cell control since they are not expected to be

affected by environmental provisions as much as green and dirty products are.

The results indicate that there is no effect on the green margin: the export of green goods relative to neutral ones does not respond to the depth of environmental provisions in China’s PTAs. The point estimate stays within a band of a few tenths of a percent per provision across every design we can construct from these data, although its precision varies with which destinations are in the sample (Appendix Table 9). The baseline uses the full firm-product-destination-year panel; a collapsed specification aggregates the same data to HS6-destination-year cells; both are replicated with the TREND provision count in place of the World Bank index, and with the DESTA index in place of the World Bank measure of general agreement depth. Three further designs replace the comparison group rather than the measurement: products are compared only within the same HS4 heading, destinations are matched on pre-treatment observables, and the untreated destinations are dropped altogether so that identification rests on differences in environmental depth among agreement partners. Across all of these the green coefficient stays inside a narrow band around zero, and it is never close to significance under wild cluster bootstrap inference (Cameron, Gelbach, and Miller, 2008) or permutation inference (Young, 2019). The null is also bounded: the bootstrap confidence interval rules out effects larger than roughly one quarter of the benchmark magnitude in the aggregate literature, on an indicative per-provision comparison.

Results on the dirty margin are less clear-cut, and we report them as such. The coefficient is negative almost everywhere, -0.4 percent per provision in the firm-level panel (about -2.6 percent for an agreement with the ASEAN chapter’s six provisions) and roughly 2.7 times larger in the collapsed panel — a gap that measures how much of the association runs through a change in which firms export rather than through reallocation inside a given firm. Whether it is statistically distinguishable from zero depends on which measure of non-environmental agreement depth is used as a control, because the two available measures differ in how much independent variation they leave in environmental depth. The test that does not depend on that choice is the permutation test, which reassigns the environmental profiles of the agreements at random across the treated destinations: it returns p -values between 0.15 and 0.38. On that evidence the dirty association cannot be distinguished from what random relabelling of the same destinations produces.

The findings show that environmental provisions in China’s PTAs between 2000 and 2015 had no measurable effect on export composition at the firm-product-destination-year level. This result, however, should not be taken as conclusive evidence that EPs in trade agreements have no effect on trade flows at all, but rather reflects the nature of the specific provisions themselves. During the period under analysis, the environmental content of China’s agreements consisted almost entirely of soft cooperation language, rather than

specific and enforceable clauses backed by dispute settlement mechanisms. The types of clauses that have been shown to produce significant trade outcomes in other contexts, such as tariff concessions on environmental goods and binding emissions or process standards, are largely absent from these agreements. In this sense, the null result is not a puzzle, but rather the expected outcome given the existing literature’s own heterogeneity results, applied to a setting where the provisions lack the specificity and enforceability that drive the effects documented elsewhere. The fact that the null result is robust to a variety of estimation designs and inference methods further supports the conclusion that the lack of effect is not due to methodological shortcomings, but rather reflects the substantive nature of the environmental provisions in China’s PTAs during this period. Moreover, the findings are informative for policy makers, as they suggest that the mere inclusion of environmental provisions in trade agreements may not be enough to achieve the desired trade outcomes if those provisions are just loose cooperation language, even in the case of the largest exporter in the world. Instead, the design and content of the provisions themselves, including specificity and enforceability, are likely to be crucial for achieving meaningful trade effects.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and the agreements. Section 4 presents the empirical strategy. Section 5 reports the results, moving from the firm-level panel to the collapsed panel, then to alternative comparison groups, the content of the agreements, and the dynamic and extensive-margin checks; the supporting robustness tables are collected in the appendix. Section 6 concludes.

2 Related Literature

A large body of evidence documents that preferential trade agreements increase trade between member countries. Using a panel gravity model that addresses PTA endogeneity, Baier and Bergstrand (2007) estimate that an FTA roughly doubles bilateral trade over a ten-year period, a finding substantially larger than earlier cross-sectional estimates — as the authors themselves note — that did not control for selection into agreements. Subsequent work shows that the overall trade effect depends on agreement characteristics: the magnitude varies with depth and enforceability rather than being a uniform response to preferential market access (Baier and Bergstrand, 2009). A survey by Freund and Ornelas (2010) argues that the distinction between shallow agreements, which primarily reduce tariffs, and deep agreements, which extend to regulatory harmonization and other behind-the-border provisions, is central to understanding their economic effects.

This distinction has been formalized through provision-level coding efforts. The World

Bank’s Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) classifies over 280 agreements into dozens of policy areas, revealing that agreements with similar headline coverage can differ dramatically in substantive content. The DESTA depth index (Dür, Bacchini, and Elsig, 2014) provides an independent coding across seven policy areas, confirming the same heterogeneity. Drawing on the World Bank Deep Trade Agreements database, Mattoo, Mulabdic, and Ruta (2022) find that deep agreements generate more trade creation and less trade diversion than shallow ones, and that some deep provisions have a public-good character that increases trade even with non-members. Taken together, these results point to a consistent conclusion: the effects of trade agreements depend critically on *what* is inside them. Whether the same holds for environmental clauses is the question this paper asks.

The interaction between environmental regulation and trade is organized around two mechanisms. The pollution haven hypothesis (Copeland and Taylor, 1994) predicts that stricter regulation erodes comparative advantage in dirty goods, shifting pollution-intensive production toward countries with weaker standards. The Porter hypothesis (Porter and Linde, 1995) predicts the opposite: well-designed regulation can trigger innovation and improve competitiveness. The empirical record on the first mechanism is mixed: while Antweiler, Copeland, and Taylor (2001) find that the income-driven technique effect of trade dominates the composition effect at the aggregate level, Levinson and Taylor (2008) instrument for abatement costs and show that industries facing the largest compliance cost increases see the largest rise in net imports — consistent with the pollution haven prediction once one addresses the econometric difficulties — chiefly endogenous abatement costs and omitted environmental regulation — that limited earlier studies. The dirty-industry classification used here draws on the pollution-intensive sector taxonomy introduced by Mani and Wheeler (1998) and Low and Yeats (1992).

Whether environmental provisions embedded in trade agreements can produce effects analogous to those of unilateral environmental regulation has been addressed by a small but growing literature. Two recent surveys reach a similar conclusion: the evidence on the effects of environmental provisions in trade agreements remains thin, and the results are highly dependent on the type of provision and the empirical setting (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2025). A few empirical studies point in the same direction.

Brandi et al. (2020) address a question closely related to the one posed in this paper. Combining the TREND environmental provision database (Morin, Pauwelyn, and Hollway, 2017) with bilateral trade flows from UN Comtrade for a panel of over 180 countries, they find that only provisions with a direct trade mechanism produce measurable effects: trade-restrictive provisions reduce dirty exports by about 5 percent of the mean; liberal provisions raise the green export share; general cooperation language shows no effect. Berger et al.

(2020), using a similar gravity framework, find that PTAs with more environmental provisions are associated with lower bilateral exports, with the effect concentrated in South-North flows. The difference between these results comes down to provision heterogeneity: Brandi et al. (2020) decompose environmental content into distinct categories, while Berger et al. (2020) use an aggregate count that combines provisions with and without a trade mechanism into a single index, which masks the heterogeneity that drives Brandi et al.’s result.

Abman, Lundberg, and Ruta (2024) exploit within-agreement variation across environmental chapters: comparing agreements with and without specific forest preservation clauses, using satellite-based deforestation data, they find that these clauses offset the increase in deforestation that typically follows PTA entry into force. The finding demonstrates that environmental provisions can produce real environmental effects — but only when they contain a specific and identifiable mechanism. On the methodological side, Cherniwchan (2017) uses within-plant US manufacturing data to show that emissions changes under NAFTA operate through individual producers’ behavior rather than market selection, demonstrating the value of micro-level data for this question.

A recent strand of work has brought the question specifically to China. Yue and Lin (2024) find that environmental provisions in China’s PTAs inhibit exports of energy-consuming products. Zhu and C. Sun (2026) use the TREND database and report that EPs reduce dirty exports and increase green exports. Zhu and C. Sun (2025) document that environmental provisions are associated with quality improvements in Chinese firms’ exports and increased imports of intermediate goods. The contrast with Zhu and C. Sun (2026), whose customs data overlap with those used here, deserves to be explicit. Their finding is identified from variation *across* firms and destinations — precisely the channel that firm-destination-year fixed effects absorb in the present design. The two sets of results are not in contradiction: Section 5.3 shows that moving from the across-firm comparison to the within-firm comparison shrinks the dirty coefficient by a factor of 2.7, a direct measure of how much of the aggregate association reflects between-firm composition rather than reallocation within the firm.

What emerges from this body of work is a consistent pattern: measurable effects on trade composition arise only when environmental provisions contain specific, enforceable clauses with an identifiable trade mechanism. General cooperation language, which characterizes the vast majority of chapters, does not produce detectable effects in studies that distinguish among provision types. This paper occupies a symmetric position relative to Brandi et al. (2020): the Chinese agreements in the period under study are dominated by exactly the kind of cooperation language that produced no effect in their decomposition. The present null is therefore the predicted outcome of the existing literature’s own heterogeneity findings, applied to a sample where the treatment lacks the specificity that drives effects documented

elsewhere.

3 Data

The analysis combines two data structures. On the treatment side, we draw on the environmental content of China’s preferential trade agreements as coded independently by the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and by TREND (Morin, Pauwelyn, and Hollway, 2017). On the outcome side, we use the universe of Chinese customs transactions for 2000–2015 at the firm–HS6–destination–year level: 45.8 million observations after excluding Hong Kong and Macao, covering 462,651 exporting firms, 236 destination economies, and between 4,450 and 4,766 HS6 product codes depending on the year. The outcome variables are the logarithm of export value, the logarithm of export quantity, and the logarithm of unit value. Products are classified as green (OECD CLEG list), dirty (pollution-intensive sectors), or neutral. Table 1 defines the main variables used throughout the analysis.

Table 1: Variable descriptions

Variable	Notation	Description
Log export value	$\ln x_{fpdt}^{val}$	Logarithm of export value (USD)
Log export quantity	$\ln x_{fpdt}^{qua}$	Logarithm of export quantity
Log unit value	$\ln x_{fpdt}^{uv}$	Log value minus log quantity
Green product	g_p	= 1 if HS6 code is in the OECD CLEG list
Dirty product	b_p	= 1 if HS6 code belongs to a pollution-intensive sector
EP depth (WB)	EP_{dt}^{WB}	Count of environmental-provision categories (World Bank)
EP count (TREND)	EP_{dt}^{TR}	Count of individual environmental clauses (TREND)
Non-env. depth	TD_{dt}	Count of non-environmental provisions (World Bank)
Log tariff	τ_{pdt}	Logarithm of (1+ applied MFN tariff rate)
Log HHI (BACI)	HHI_{pdt}	Log Herfindahl index of destination-product concentration

Notes: Subscripts: f = firm, p = HS6 product, d = destination, t = year. Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

3.1 Chinese PTAs and their environmental content

Between 2000 and 2015, China signed preferential agreements covering 25 destination economies, of which 23 enter the main estimation sample after excluding Hong Kong and Macao (Table 2); Figure 2 maps them. Timor-Leste is coded as an ASEAN–China party in the agreement lists although it is not one; it accounts for 0.02 percent of observations and its exclusion

moves no coefficient beyond the sixth decimal (Table 9). Three features of this treatment landscape shape everything that follows.

Table 2: Chinese PTAs in force, 2000–2015

Agreement	Destinations	In force	WB EP	TREND EP
Bangkok Agr./APTA	Bangladesh, India, South Korea, Laos, Sri Lanka	2002	1	1
ASEAN–China	Brunei, Cambodia, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Vietnam	2005	6	4
China–Chile	Chile	2006	5	30
China–Pakistan	Pakistan	2007	3	12
China–New Zealand	New Zealand	2008	7	53
China–Singapore	Singapore	2009	7	15
China–Peru	Peru	2010	12	47
China–Costa Rica	Costa Rica	2011	4	50
China–Iceland	Iceland	2014	6	28
China–Switzerland	Switzerland	2014	14	53
China–Australia	Australia	2015	3	24
China–South Korea	South Korea	2015	17	72
<i>Excluded from main specification</i>				
CEPA Hong Kong	Hong Kong	2003	4	5
CEPA Macao	Macao	2003	5	5
Total treated destinations (excl. HK–MO)				23

Notes: Entry-into-force years from the World Bank Deep Trade Agreements database and TREND. A destination is coded as treated from the calendar year of entry into force. Seven agreements enter in the second half of the year (Chile 1 October 2006, Pakistan 1 July 2007, New Zealand 1 October 2008, Costa Rica 1 August 2011, Iceland and Switzerland 1 July 2014, Australia and Korea 20 December 2015). WB EP = count of environmental-provision categories (World Bank); TREND EP = count of individual environmental clauses (TREND). Values shown are the maximum observed over the sample period. Three destinations experience within-country increases in EP depth: Laos (Bangkok 2002 → ASEAN 2005), Singapore (ASEAN 2005 → bilateral 2009), and South Korea (Bangkok 2002 → bilateral 2015). Laos is listed under the Bangkok Agreement, its first agreement; it is also a party to ASEAN–China from 2005, so eleven destinations are ASEAN–China parties in total. Source: World Bank DTA Database; TREND Database.

First, the 23 treated destinations (listed in Appendix Table 15) correspond to only 13 distinct EP profiles—combinations of depth and entry timing. Ten destinations are parties to the ASEAN–China agreement; Laos also entered through the Bangkok Agreement and Singapore also signed a separate bilateral, leaving nine destinations that share the same ASEAN-only profile. Standard errors clustered by destination and the permutation inference described in Section 4.2 account for this structure.

Second, EP content is fixed at signature and almost never changes afterwards. Only three destinations (Laos, Singapore, and South Korea) exhibit within-country variation over time, and in each case depth rises, never falls. There is no treatment reversal in the sample.

The last two agreements deserve a caveat. Australia and Korea entered into force on 20 December 2015, the final year of the panel, so their treated year contains eleven days of exposure; Korea’s jump from one to seventeen provisions, the deepest chapter in the sample, is identified from that year alone. Appendix M re-estimates the main specification coding these two destinations, and all second-half entrants, as untreated in their entry year.

Third, Hong Kong and Macao (CEPA, 2003) are entrepôt economies. They account for 24.4% of treated observations and 50.1% of treated export value, but a large fraction of what the customs data record as Chinese exports to these destinations are goods transshipped and re-exported to third countries at a markup (Feenstra and Hanson, 2004). Including them would mix the demand of the true final destination with the bilateral fixed effects designed to absorb it, since the recorded destination and the actual final market differ systematically.¹

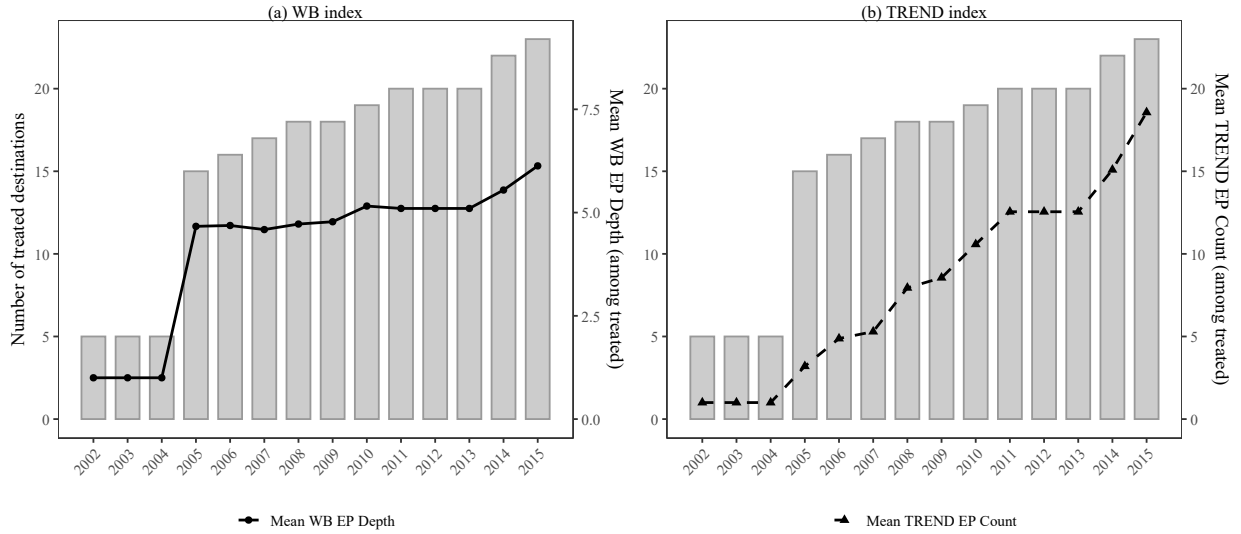
Entry into the treated group is concentrated in two large cohorts (2002 and the 2005 ASEAN accession); most subsequent years add one destination at a time. Of the 236 destinations in the panel, 23 are ever treated and 213 are never treated within the sample window.

EP content is measured with two independent codings. The World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) yields EP_{dt}^{WB} , the count of environmental-provision categories in force with destination d at time t , and a non-environmental depth control TD_{dt} (total non-environmental provisions). TREND (TRade and ENvironment Database; Morin, Pauwelyn, and Hollway, 2017) yields EP_{dt}^{TR} , a count of individual environmental clauses, and sub-indices covering green market access, enforcement stringency, and hard versus soft provisions. Using two codings independently constructed from different frameworks guards against the idiosyncrasies of either; as shown below, they agree on the main findings.

Figure 1 traces the expansion of China’s PTA network and the evolution of EP content over the sample period. The early agreements (Bangkok 2002, ASEAN 2005) contain minimal environmental language; substantive environmental chapters appear only from New Zealand (2008) onward, with the deepest content in the Switzerland and South Korea agreements (2014–2015).

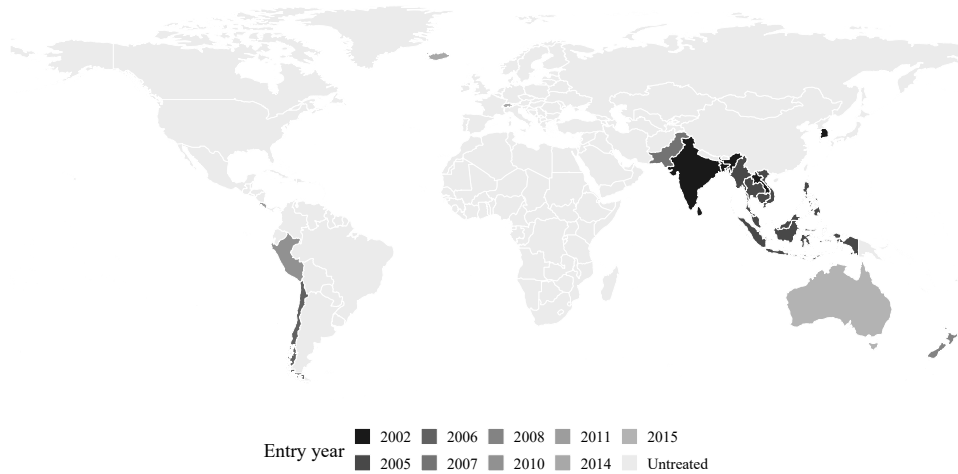
¹Results including Hong Kong and Macao are reported in Appendix J. The green coefficient is unchanged; the collapsed-panel dirty coefficient roughly doubles, and dropping Hong Kong alone reverses almost all of the increase, which is what one expects if part of it reflects transshipment rather than final demand.

Figure 1: Expansion of China's PTA network and EP content, 2000–2015



Notes: Bars show the number of treated destinations (excl. HK–MO) with positive EP depth in each year. Lines show the mean EP index value among treated destinations: WB EP Depth (solid circles, left panel) and TREND EP Count (dashed triangles, right panel). The two indices are plotted on separate axes because they differ in scale by a factor of approximately three by 2015. Source: World Bank DTA Database; TREND Database.

Figure 2: Geographic coverage of Chinese PTAs, 2000–2015



Notes: Shaded countries are destinations covered by a Chinese PTA with environmental provisions during the sample period. Shading indicates the year of entry into force. Hong Kong and Macao (excluded from the main specification) are not shown. Source: World Bank DTA Database.

3.2 Customs data and product classification

The trade data are the universe of Chinese export customs transactions for 2000–2015, sourced from the Chinese Customs Office. Transactions are aggregated to the firm–HS6–destination–year level, yielding 49.2 million observations (45.8 million once Hong Kong and Macao are excluded), 462,651 exporting firms, 236 destinations, and between 4,450 and 4,766 HS6 product codes depending on the year. Three outcome variables are used throughout: the logarithm of export value ($\ln x_{f\text{pdt}}^{\text{val}}$), the logarithm of export quantity ($\ln x_{f\text{pdt}}^{\text{qua}}$), and the logarithm of unit value ($\ln x_{f\text{pdt}}^{\text{uv}}$, defined as value divided by quantity). Export value is the primary outcome; quantity and unit value serve as decomposition checks.

Green products are the 248 HS6 codes of the OECD Combined List of Environmental Goods (CLEG), as classified in Sauvage (2014). The customs panel uses the HS1996 vintage while the OECD list is defined in HS2012, so the list was harmonized. Of the 248 codes, 246 have a one-to-one mapping to HS1996 and are translated directly. The remaining two (codes 871411 and 871419, which correspond to a single HS1996 heading split into two sub-codes in HS2012) are retained at the original HS1996 level; these two codes jointly account for a negligible share of green-goods trade, so no information of consequence is lost.²

Dirty products belong to the classic pollution-intensive sectors of Mani and Wheeler (1998) and Low and Yeats (1992): pulp and paper, industrial chemicals, petroleum refining, iron and steel, and non-ferrous metals, with cement added in an extended variant.³

The remaining HS6 codes form the neutral comparison group. Summary statistics appear in Table 3.

Figure 3 plots the share of green and dirty products in total export value over time, separately for treated and untreated destinations. For green products the two groups track each other closely throughout, with no visible divergence after agreements enter into force. The dirty share is about twice as high toward treated destinations and rises faster between 2002 and 2008, which reflects the product mix of the early partners (India, ASEAN) rather than a response to environmental content: the level difference is absorbed by the destination fixed effects, and the pre-treatment coefficients of the event study (Figure 4) are flat.

²As a check on the classification itself, the specification was re-estimated restricting green products to the 54 codes of the APEC Environmental Goods List (Sauvage, 2014), on which there is explicit multilateral political consensus and which form a complete subset of the OECD list. With the WB index the green coefficient flips sign to +0.0050 (s.e. 0.0127, $p = 0.69$); with TREND it moves from +0.0018 to +0.0032 ($p = 0.13$). Both remain indistinguishable from zero, with standard errors roughly double, as expected from restricting to 22% of the original green sample. Appendix Table 32 reports the full comparison.

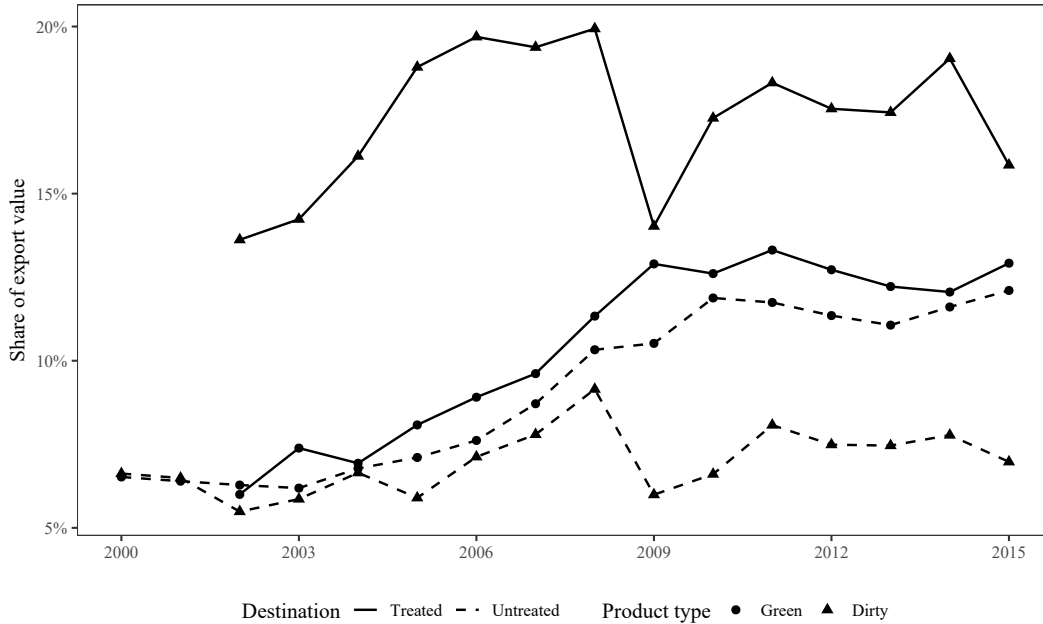
³Products are mapped from ISIC Rev.2 to HS1996 using the official WITS/UNSD concordance, yielding 1,139 dirty HS6 codes. Seventeen codes that appear in both lists are assigned to the green category; the two lists are mutually exclusive.

Table 3: Summary statistics (full panel, excl. HK-MO)

Variable	N	Mean	Median	Std. Dev.	Range
$\ln x_{fpdt}^{val}$	45,781,181	9.264	9.360	2.546	[0, 23.3]
$\ln x_{fpdt}^{qua}$	45,470,053	7.312	7.595	3.213	[-6.9, 25.7]
$\ln x_{fpdt}^{uv}$	45,470,040	1.986	1.609	2.306	[-13.0, 20.4]
g_p (green)	45,781,211	0.115	0	0.319	{0, 1}
b_p (dirty)	45,781,211	0.070	0	0.255	{0, 1}
EP_{dt}^{WB}	45,781,211	0.966	0	2.383	[0, 17]
EP_{dt}^{TR}	45,781,211	2.051	0	8.165	[0, 72]
τ_{pdt}	41,117,073	1.658	1.792	1.027	[0, 7.3]
HHI_{pdt}	45,306,336	-1.540	-1.610	0.592	[-3.3, 0]
<i>Collapsed panel</i>					
$\bar{y}_{pdt}$ (collapsed)	3,773,498	8.974	9.029	1.957	[0, 20.7]
Cell count n_{pdt}	3,773,498	12.13	3	43.64	[1, 8,946]

Notes: 45,781,211 firm-HS6-destination-year observations after excluding Hong Kong and Macao. Observation counts differ across variables due to missing tariff data and quantity/unit-value availability. Thirty observations with zero recorded export value have no log export value. Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Figure 3: Green and dirty export shares by treatment status, 2000–2015



Notes: Share of export value accounted for by green and dirty products, computed separately for destinations with positive EP depth (treated) and destinations with no PTA or no EP content (untreated) in each year. Excludes Hong Kong and Macao. Source: Chinese Customs Office.

3.3 Estimation samples

The analysis rests on four estimation objects built from the same underlying panel; their sizes and fixed-effect structures are summarized in Table 4.

Table 4: Descriptive statistics

<i>Full firm-level panel (excl. HK-MO)</i>	
Observations (firm-HS6-dest-year)	45,781,211
Exporting firms	462,651
HS6 products	4,450-4,766
Destinations	236
of which treated (EP depth > 0)	23
Distinct EP profiles (depth $\times$ timing)	13
Share of observations: green products	11.5%
Share of observations: dirty products	7.0%
Share of observations under an EP agreement	20.3%
<i>Collapsed panel (HS6-dest-year)</i>	
Cells	3,773,498
of which green / dirty	8.4% / 14.0%
Estimation sample (post-singleton)	3,681,023
<i>PPML zero-filled grid (HS6-dest-year, excl. HK-MO)</i>	
Cells	8,179,904

Notes: Hong Kong and Macao excluded throughout (they account for 24.4% of treated observations and 50.1% of treated export value). Product classification as described in the text. Source: Chinese Customs Office.

The green and dirty shares differ between the two panels because collapsing changes the unit of observation from the firm to the cell: green products are exported by more firms per product-destination cell, so their weight falls from 11.5% of observations to 8.4% of cells, while dirty products have fewer exporting firms per cell and their weight rises from 7.0% to 14.0%. The two panels contain the same underlying transactions.

The *full firm-level panel* is the main estimation object, with 45.8 million observations at the firm-HS6-destination-year level. The outcome variables are $\ln x_{fpdt}^{val}$, $\ln x_{fpdt}^{qua}$, and $\ln x_{fpdt}^{uv}$; the main fixed effects are firm-product-destination (θ_{fpd}), firm-destination-year (θ_{fdt}), and product-year (θ_{pt}). Standard errors are clustered by destination.

The *collapsed panel* aggregates the firm dimension, yielding 3.77 million HS6-destination-year cells. The outcome in each cell is the mean of log exports across the underlying firm-level observations, and regressions are weighted by the cell count (the number of firm-level observations in each cell). Fixed effects are product-destination (pd), destination-year (dt), and product-year (pt). The collapsed panel serves two purposes: it is a computational

workhorse for the bootstrap and permutation inference battery, and it provides an internal cross-check against the full panel.⁴ The two panels agree on the green margin; Section 5.3 explains why they diverge on the dirty margin.

The *zero-filled PPML grid* (8.2 million cells) is built from the full panel by first aggregating to HS6–destination–year cells and then adding product–destination pairs with no recorded trade in a given year, conditional on the pair having at least one positive flow elsewhere in the sample period.⁵

Because the triple-difference compares green and dirty products to neutral ones, the credibility of the comparison group can be probed directly. Four control-group subsamples, each tightening the comparison in the direction of a specific threat, are summarized in Table 5 and described below. The prefix C- denotes the control-group variant.

Table 5: Control-group subsamples

Subsample	Threat addressed	Obs.
C-prod-HS4 <i>neutral products in the same HS4 family as a green code</i>	Neutral products differ from green and dirty ones	3.8M
CEM destinations <i>destinations matched on year-2000 GDP p.c., growth, MFN tariff</i>	Selection of partner countries	14.0M
Deep vs. shallow <i>PTA partners only, split at median EP depth</i>	The “any-agreement” contrast it-self	5.3M

Notes: All observation counts refer to the full firm-level panel (excl. HK–MO). Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

C-prod-HS4 restricts the neutral comparison group to products belonging to the same HS4 family as a green code (106 HS4 families, covering 20.5% of observations). This guards against the concern that neutral products are too dissimilar from green ones to serve as a valid counterfactual. It should be read alongside C-overlap: multi-product firms may reallocate

⁴With 45.8 million observations and three sets of high-dimensional fixed effects, the full panel is computationally intensive, and the bootstrap requires several hours per specification. The collapsed panel reduces dimensionality by a factor of twelve, making the full inference battery feasible on a single machine. The equivalence between the two implementations is verified algebraically and numerically in Appendix A.

⁵This conditional zero-fill — rather than a complete cross-join of all products, destinations, and years — keeps the grid computationally tractable while capturing the relevant extensive margin. A zero-fill at the firm–product–destination–year level would generate billions of rows, making estimation computationally infeasible; a PPML without zero-fill would miss the extensive margin entirely, since the estimator’s advantage lies precisely in the information contained in the zeros.

toward core products when trade costs fall (Eckel and Neary, 2010), potentially displacing neutral products in the same HS4 family and biasing the contrast.

C-overlap retains only products exported to both treated and untreated destinations (98.5% of all HS6 codes, 21.5 million observations). This ensures that the identifying variation does not rely on products traded exclusively with one group of destinations, guarding against the possibility that multi-product firms reallocate across products within the same HS4 family when trade costs fall ().⁶

CEM destinations applies coarsened exact matching (Iacus, King, and Porro, 2012) at the destination level, matching treated destinations to untreated ones on year-2000 GDP per capita, GDP growth, and MFN tariff levels. The multivariate L_1 imbalance falls from 0.68 before matching to 0.43 on the matched sample. The resulting sample contains 19 treated and 40 control destinations (13,992,396 observations after iterative singleton removal) and addresses the concern that treated and untreated destinations differ systematically on dimensions correlated with export composition.

Deep vs. shallow drops all untreated destinations and splits the remaining PTA partners at the median EP depth (median computed over the 25 PTA partners including Hong Kong and Macao; 16 deep and 9 shallow, of which 16 and 7 remain in the baseline sample; 5.3 million observations). This design asks whether composition differs between deep and shallow environmental chapters, conditional on having an agreement at all, and addresses the concern that any observed effect reflects the agreement rather than its environmental content.

A genuine composition effect should survive every change to the comparison group; an artifact will move when the comparison changes.

4 Empirical Strategy

Three challenges shape the identification strategy: countries select into agreements, environmental and overall agreement depth correlate strongly, and the number of independent treatment units is small.

A starting point would be to estimate the level effect of environmental provisions on trade—regressing log exports on environmental depth with fixed effects. A structural argument shows why this is uninformative in this setting.

The argument is independent of the empirical results. Any fixed-effects structure that seriously addresses bilateral confounders requires a destination–year intercept, which absorbs

⁶In practice the restriction is close to non-binding: it removes 314 of the 21,519,511 observations in the estimation sample, and the estimates it produces are indistinguishable from the baseline to the fourth decimal. For that reason it is reported in the robustness battery of Appendix K rather than alongside the three subsamples that genuinely change the comparison group in Section 5.4.

all variation at the (d, t) level—including environmental depth itself. With only 13 distinct EP profiles and near-zero within-destination change over time, “environmental depth” and “having a deep agreement” are empirically indistinguishable. The interaction $EP_{dt} \times g_p$ survives the destination–year intercept because the product dimension varies *within* the (d, t) cell. This is why the specification is built on composition—the differential response of green and dirty products relative to neutral ones—rather than on levels.

This design cannot show that EP-heavy agreements affected aggregate trade levels — that effect is confounded with the agreement itself. The evidence speaks only to the composition margin.

4.1 The triple-difference specification

The question is whether environmental depth shifts the gap between green (or dirty) and neutral log exports among treated destinations.⁷ OLS gives equal weight to each observation, so the coefficient is a per-observation average, not a per-firm or per-destination average.⁸ The main specification is

$$\ln x_{fpdt} = \beta_1 EP_{dt} \times g_p + \beta_2 EP_{dt} \times b_p + \gamma_1 TD_{dt} \times g_p + \gamma_2 TD_{dt} \times b_p + \theta_{fpd} + \theta_{fdt} + \theta_{pt} + \varepsilon_{fpdt}, \quad (1)$$

where f indexes firms, p HS6 products, d destinations, and t years; g_p and b_p are the green and dirty product indicators defined in Section 3; EP_{dt} is environmental depth and TD_{dt} non-environmental agreement depth. Environmental depth enters the regression only through its interactions with g_p and b_p , not on its own: the level effect of EP_{dt} is absorbed by θ_{fdt} , and what remains is the differential response of green and dirty products relative to neutral ones.

The three sets of fixed effects absorb different sources of confounding variation, and the choice of which to include is dictated by the structure of the treatment.

The firm–product–destination fixed effects θ_{fpd} absorb time-invariant features of each firm–product–market relationship—for example, a firm’s long-standing contract with a South Korean buyer of solar panels, or an established network for exporting steel to a particular ASEAN destination.

The firm–destination–year fixed effects θ_{fdt} are what drives identification. They absorb all time-varying bilateral factors — entry into new markets, aggregate demand shifts, exchange

⁷In potential-outcomes terms, the target can be written as $\beta_1 = E[(y_{fpgdt}(1) - y_{fpgdt}(0)) - (y_{fpndt}(1) - y_{fpndt}(0)) \mid d, t \text{ treated}]$, where $y(1)$ and $y(0)$ denote log exports of green product g versus neutral product n under deeper and unchanged environmental content respectively, holding the agreement fixed.

⁸A firm exporting 100 products receives 100 times the weight of a firm exporting one product. The coefficient therefore reflects the composition of the estimation sample rather than a simple average across firms or destinations.

rate movements — that affect a firm’s exports to a destination uniformly across products. The agreement dummy, overall agreement depth, destination demand shocks, and the non-random selection of destinations into agreements all drop out. Identification comes from comparing green and dirty products to neutral ones *within* the same cell as environmental content varies across destinations and over time.

The product–year fixed effects θ_{pt} absorb global product-level shocks common to all destinations—for instance, the worldwide expansion of the solar panel market in the late 2000s, or global fluctuations in steel prices—so that these shocks do not contaminate the composition comparison.⁹

The three-way fixed-effect structure absorbs a large number of singletons — observations whose identifying variation is fully accounted for by at least one fixed effect. Iterative singleton removal (Correia, 2017) reduces the estimation sample from 45.8 million raw observations to 21,519,511 in the full panel (a 53% reduction) and from 3,773,498 cells to 3,681,023 in the collapsed panel.

Selection into agreements is not eliminated but substantially narrowed: $\theta_{f dt}$ absorbs the level effect, so selection contaminates β_1 only through differential pre-trends in green versus neutral exports across destinations. Appendix H tests this directly with destination-specific trends and finds no change in the green coefficient.

The omitted category is neutral products. β_1 measures how green products move *relative to neutral ones* as environmental depth increases; β_2 measures the same contrast for dirty products. Each coefficient is already the difference from the neutral baseline. The green–dirty differential ($\beta_1 - \beta_2$) is a derived quantity, not the parameter the regression identifies, and should not be read as such.

The same World Bank non-environmental depth control is used in both the WB and the TREND specifications: TREND does not code non-environmental provisions, so no TREND-based analogue exists.

The non-environmental depth interactions ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) separate environmental from general agreement content. TD_{dt} counts all non-environmental provisions, but not all of these are directly relevant to green or dirty trade; it is therefore an imperfect proxy for the non-environmental component of agreement depth that might independently affect export composition. An imperfectly measured control under-controls: β_1 absorbs whatever share of general-depth variation the proxy fails to capture.¹⁰ Rather than assert a direction,

⁹Firm–product–year fixed effects would be computationally infeasible given the panel size. Product–year effects are sufficient because firm–product shocks that do not vary across destinations are already absorbed by the combination of $\theta_{f pd}$ and $\theta_{f dt}$.

¹⁰If the non-environmental depth control imperfectly measures the true confounding depth, $\hat{\beta}_1$ absorbs the part of general-depth variation the proxy misses. Because EP and total depth correlate at $\rho = 0.91$, this omitted component is positively associated with EP, so the bias pushes $\hat{\beta}_1$ upward. With a null green

Appendix L reports the green coefficient under four alternative depth controls and shows that the point estimate moves within a band narrower than one standard error (Table 29).

Collinearity between EP depth and non-environmental depth is substantial: the two correlate at $\rho = 0.91$ across treated destination-years — and more tightly still once the fixed effects are partialled out, as Section 5.2 shows — and the variance inflation factor of EP depth regressed on non-environmental depth across treated destination-years is 5.8.¹¹ The TREND count is less collinear with non-environmental depth ($\rho = 0.50$), which is one reason the WB confidence intervals are wider. That two codings with such different degrees of overlap return the same null is reassuring, since a result driven by the inability to separate environmental from general depth would be expected to differ between them.

Standard errors are clustered by destination. EP depth varies across destinations and years but changes in only three destinations during the sample period, making treatment effectively destination-level. Clustering by destination treats all observations from the same country as a single cluster, which is the appropriate level when the treatment is in practice a destination-level assignment (Abadie et al., 2023).

A further identification concern works in a conservative direction. If deeper environmental chapters tend to be signed where green demand is already rising, EP_{dt} is positively correlated with an omitted destination-specific trend, which would bias β_1 positively. With a null coefficient, any upward bias from this channel implies the true effect is zero or negative—the null is a conservative rather than a lenient reading of the data.

Because environmental depth is a continuous dose switching on at staggered dates, the coefficient is a weighted average of dose- and cohort-specific composition responses, not a simple average treatment effect. For the green margin the dose-specific estimates of Appendix N are all close to zero, so the weighting is immaterial; for the dirty margin they are not, and the linear coefficient should be read as a summary of a contrast between the medium-depth ASEAN cohort and the rest rather than as a per-provision slope.¹²

coefficient, any upward bias implies the true effect is zero or negative — the null is a conservative reading of the data.

¹¹The Variance Inflation Factor measures how much collinearity inflates the variance of the estimated coefficient relative to an orthogonal design. Here it is computed on the two depth variables at the destination-year level, not on the interacted regressors of the estimating equation. A VIF of 5.8 is somewhat above the maximum of 4.6 that Brandi et al. (2020) report for their provision-type variables on a panel of 680 agreements.

¹²Under continuous treatment entering at staggered dates, the TWFE estimator produces a weighted average of dose- and cohort-specific effects whose weights need not be convex (Callaway, Goodman-Bacon, and Sant’Anna, 2024; Chaisemartin and D’Haultfoeuille, 2020; Goodman-Bacon, 2021). Two features limit what this costs in this setting. First, a weighted average of near-zero effects stays near zero unless the underlying effects are large and of opposite signs, of which the sub-index and leave-one-out evidence gives no indication. Second, the permutation test is agnostic about weighting: it asks whether the observed coefficient—whatever weighted average it represents—is unusual relative to placebos.

4.2 Inference with few treated clusters

Of the destination clusters in the estimation sample, only 23 are treated, corresponding to 13 distinct EP profiles. With so few treated clusters, asymptotic cluster-robust standard errors are unreliable: the finite-sample approximation to the t -distribution is too optimistic, and p -values systematically over-reject the null (Cameron, Gelbach, and Miller, 2008; MacKinnon and Webb, 2017). Every headline estimate is therefore subjected to two additional tests.

Wild cluster bootstrap. The wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008; Roodman et al., 2019) provides inference that does not rely on asymptotic critical values and is appropriate for settings with few clusters. In both panels, fixed effects are partialled out via the Frisch–Waugh theorem before the bootstrap is applied, because the bootstrap implementation requires a single set of absorbed effects. Point estimates are identical to those from the direct estimation; only computational cost differs.¹³

Permutation inference. The permutation test reshuffles EP profiles across the 23 treated destinations, testing whether the specific EP content — not the agreement itself — matters for composition.¹⁴ With 1,000 replications, the test asks whether the observed coefficient is unusual relative to the placebo distribution.

Because depth and timing are permuted jointly, the test cannot separate the effect of environmental content from the effect of entry timing. The whole time path of each profile moves with it, including entry timing and the non-environmental depth control, so the null hypothesis is that, given the set of destinations that ever sign, it does not matter which of them received which profile.¹⁵

¹³Partialling out introduces two minor approximations. First, the bootstrap treats all fixed effects as if nested within the destination cluster. In the full panel this is nearly true: firm–product–destination and firm–destination–year are each nested within a destination; only product–year is not. In the collapsed panel the nesting is less complete (only destination–year is strictly nested), so the approximation is somewhat less tight. Second, the small-sample adjustment sees only the residualized regressors rather than the regressors plus the absorbed fixed effects. Neither approximation has practical relevance here: the conclusions are insensitive to the choice of panel, and the collapsed-panel dirty coefficient has a bootstrap p -value of 0.07 and the full-panel one of 0.19 regardless of which approximation applies.

¹⁴The test operates only among treated destinations because permuting between treated and untreated destinations would test a different hypothesis (the effect of the agreement as such), which is unidentifiable as shown in Section 4.

¹⁵Estimates on the collapsed and destination-level panels were produced twice, in STATA (`reghdfe`, Correia, 2017; `boottest`, Roodman et al., 2019; `ppmlhdfe`, Correia, Guimarães, and Zylkin, 2020; `eventstudyinteract`, L. Sun and Abraham, 2021) and in R (`fixest`, `fwildclusterboot`); across 44 result files the two implementations agree on every point estimate to at least seven significant digits. Estimates on the 45.8-million-row firm-level panel exist only in STATA: `fixest` cannot complete them on the available hardware. They are validated instead by an algebraic identity—the firm-level panel estimated with $pd+dt+pt$ fixed effects reproduces the weighted collapsed-panel estimate to nine significant digits (Appendix A)—which checks the data construction of the two pipelines against each other. All reported numbers are the STATA ones. Two quantities legitimately differ between implementations and are discussed where they arise: bootstrap and permutation p -values, which carry Monte Carlo error, and the Sun–Abraham standard errors of Appendix G,

Bootstrap and permutation p -values accompany every estimate in Section 5; the complete set, across both panels, both indices and all four sample–depth variants, is collected in Appendix C (Tables 16 and 17).

5 Results

5.1 Fixed effects and the reference specification

Before reading a single coefficient it is worth establishing that the fixed-effect structure of equation (1) earns its cost. Table 6 estimates the same composition interactions on the same panel under five sets of absorbed effects, moving from the sparsest to the specification used in the rest of this section. The dependent variable is log export value throughout, so the columns differ only in what they hold fixed.

The first four columns tell a coherent and misleading story. The World Bank dirty interaction is positive and significant at +0.0064 in column (4), and the TREND green interaction is positive and significant at +0.0018 in the same column; in column (3) both are again positive and significant at the ten percent level or better. Read on their own, these say that agreements with deeper environmental chapters saw both green and dirty exports grow relative to neutral ones — which is not a composition effect at all, since the two ends move the same way.

What the four columns have in common is the explanation. None of them contains a fixed effect with both a destination and a time dimension. Chinese exports to a partner grow around the entry into force of an agreement for reasons that have nothing to do with its environmental chapter: tariffs fall, the partner’s income rises, the trading relationship deepens. That growth is a destination–time phenomenon, and with no destination–time absorber it has nowhere to go except onto whichever interaction is correlated with agreement timing — which is exactly what the EP interactions are, since the destinations with environmental content are the destinations with agreements.

Column (5) adds firm–destination–year effects and the picture changes completely. The World Bank dirty coefficient goes from +0.0064 to -0.0044 , the TREND green coefficient from +0.0018 to -0.0001 , and the significance disappears with the sign. We cannot say which destination-specific shock was driving the earlier columns, only that it had a destination–time structure, because absorbing that structure is what removes it. This is the empirical counterpart to the argument made on structural grounds in Section 4, and it is why the

which reflect a genuine methodological difference between `eventstudyinteract` and `fixest::sunab` (see Appendix G).

saturated specification is not one defensible choice among five but the only one in the sequence that removes a confounder visible in the data.

Column (5) is the specification of Table 7 and of everything that follows. Appendix E reports the same ladder for quantity and unit value, where the pattern repeats.

Table 6: Fixed-effects structure and the composition interactions (full panel, log export value)

	(1)	(2)	(3)	(4)	(5)
$EP_{dt}^{WB} \times g_p$	0.0018 (0.0050)	0.0012 (0.0027)	0.0060 (0.0050)	0.0050 (0.0037)	−0.0023 (0.0039)
$EP_{dt}^{WB} \times b_p$	−0.0037 (0.0150)	−0.0015 (0.0027)	0.0058* (0.0031)	0.0064** (0.0027)	−0.0044* (0.0022)
$EP_{dt}^{TR} \times g_p$	−0.0013 (0.0010)	0.0010 (0.0009)	0.0021** (0.0009)	0.0018** (0.0008)	−0.0001 (0.0010)
$EP_{dt}^{TR} \times b_p$	−0.0049*** (0.0015)	0.00003 (0.0007)	0.0002 (0.0011)	0.0009 (0.0008)	−0.0009 (0.0006)
θ_{fpd}	✓		✓	✓	✓
θ_{fpt}		✓	✓		
θ_{pd}		✓			
θ_{pt}				✓	✓
θ_t	✓				
θ_{fdt}					✓
Absorbs destination×time?	No	No	No	No	Yes
Observations	29,477,365	35,607,036	22,927,402	29,473,145	21,519,511
Clusters (d)	231	232	231	231	225

Notes: Each column is a separate estimation on the full firm-level panel with a different set of absorbed fixed effects; the dependent variable is log export value throughout. Column (5) is the reference specification of Table 7. Columns (1)–(4) differ in observation count because fewer singletons are removed under less saturated structures; none contains a fixed effect with both a destination and a time dimension. Column (5) adds θ_{fdt} . Hong Kong and Macao excluded. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Standard errors clustered at the destination level in parentheses. Quantity and unit-value outcomes in Appendix Tables 19 and 20. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

5.2 The full panel

Table 7 reports the estimates on the full firm–product–destination–year panel. There are twelve coefficients of interest in it, and they are easier to read across than down. Columns (1)–(3) measure environmental content with the World Bank depth index and columns (4)–(6) with the TREND provision count; within each block the outcome is log export value, then log quantity, then log unit value. Panel A controls for non-environmental World Bank depth and Panel B for the independently coded DESTA index (Dür, Baccini, and Elsig, 2014). The green and dirty interactions are estimated jointly and should be read together: the design

asks whether an environmental chapter tilts a firm’s product mix, and a tilt has two ends.

Only one of those ends moves. On the green margin the coefficient is indistinguishable from zero in every cell. Export value gives -0.0023 (s.e. 0.0039) with the World Bank index and -0.0001 (0.0010) with TREND; switching the depth control to DESTA changes the third decimal and nothing else. The joint F -test on all four composition interactions returns $p = 0.31$ (WB) and $p = 0.71$ (TREND) in Panel A, and $p = 0.18$ and $p = 0.26$ in Panel B. Quantity and unit value add nothing: across the eight green cells in columns (2), (3), (5) and (6) no coefficient comes close to significance. That rules out the one reading a value regression alone cannot dismiss — that a volume response and a price response are offsetting each other inside the value aggregate. Neither channel moves. If Chinese firms shifted toward environmental goods when an agreement with environmental content entered into force, the shift is not visible here at any margin.

The dirty coefficient does move, and it is worth being precise about how much. In Panel A it is -0.0044 on export value with an asymptotic p -value of 0.052 , and -0.0009 with $p = 0.15$ under TREND. In Panel B the same two coefficients are -0.0056 ($p = 0.039$) and -0.0014 ($p = 0.048$). The sign is stable, the magnitude is stable, and the asymptotic verdict depends on which measure of non-environmental agreement depth sits in the regression. That dependence has a mechanical explanation. Within destinations and years, environmental depth correlates 0.959 with World Bank non-environmental depth and 0.891 with DESTA depth.¹⁶ The World Bank control absorbs almost the same variation as the regressor of interest, so little independent variation is left and the standard error is wide; DESTA overlaps less, leaves more, and the standard error narrows. Nothing about the estimated effect changes between the panels — only how much of it the design can see.

This is the point at which asymptotic p -values stop being informative, for the reasons set out in Section 4.2: with 23 treated destinations and 13 distinct agreement profiles, cluster-robust inference over-rejects. The row labelled “Bootstrap p ” in this and every subsequent table reports the wild cluster bootstrap with $B = 9,999$ replications, and it is the reference inference throughout. The asymptotic figures are kept alongside it only so that the gap between the two remains visible.

The gap is large exactly where it matters. On the green margin the bootstrap changes nothing: p moves from 0.57 to 0.69 (WB) and from 0.92 to 0.93 (TREND), and the null is a null either way. On the dirty margin under the World Bank control it changes the

¹⁶Correlations of the two depth measures after removing destination and year means, computed on the treated destination-years (223 with the World Bank control, 212 with DESTA, which does not cover Timor-Leste). They are higher than the raw correlation of $\rho = 0.91$ because the demeaning removes the variation the two measures share across destinations. They are a property of the treatment variables, not of the interacted regressors of equation (1).

verdict, taking p from 0.052 to 0.19 and from 0.15 to 0.18. Under DESTA it does not: 0.039 becomes 0.049 and 0.048 becomes 0.069. So the dirty coefficient survives robust inference under one depth control and not under the other, and it does so on both codings at once, which at least means the two databases are telling the same story. The place where it is strongest is quantity: under DESTA the World Bank interaction is -0.0070 on log quantity with a bootstrap p of 0.035, larger and better determined than on value, while the unit value coefficient is $+0.0004$ and nowhere near significance. Whatever this is, it is a shipment volume and not a price. Section 5.3 brings to bear the one test that does not depend on the choice of depth control.

The sign, at least, is where the recent literature on China would put it. Yue and Lin (2024) find that environmental provisions in China’s PTAs inhibit exports of energy-consuming products, and Zhu and C. Sun (2026), working with customs data that overlap with those used here, report that environmental provisions reduce dirty exports and raise green ones. On dirty the present estimates agree in sign and are considerably smaller; on green they do not agree at all. The difference is not mysterious, and Section 5.3 measures it: their identification comes from variation across firms and destinations, which is precisely the variation that θ_{fdt} removes here. Zhu and C. Sun (2025) report that environmental provisions are associated with quality upgrading in Chinese firms’ exports; unit value is a coarse proxy for quality and column (3) is null, but the outcome and the sample differ enough that this should not be read as a test of their result.

A null is only as informative as the interval around it. The 95% wild cluster bootstrap confidence interval for the green interaction runs from -0.0353 to $+0.0355$ per World Bank provision, roughly ± 3.5 percent. Since one standard deviation of World Bank EP depth in the estimation sample is about 2.4 provisions, that bound implies at most an eight to nine percent change in green relative to neutral exports for a one-standard-deviation increase in environmental content. Stated per agreement rather than per provision, the bound is looser: for the six provisions of the ASEAN chapter it is about ± 21 percent, and for the sixteen-provision jump of the Korea agreement it exceeds ± 40 percent. The null is bounded at the provision level, not at the level of a deep agreement. The corresponding asymptotic interval, $[-0.0100, +0.0055]$, is roughly four and a half times narrower; reporting that width would claim a precision Section 4.2 has already declared unwarranted. The statement this paper can defend is therefore not “there is no effect” but “any effect is smaller than about three percent per provision”, which is a bound, not an absence. Appendix Table 37 reports the corresponding minimum detectable effects on the collapsed panel, where the standard errors are larger and the bounds correspondingly looser.

Table 7: Full panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1)	(2)	(3)	(4)	(5)	(6)
	$\ln x_{fpdt}^{val}$	$\ln x_{fpdt}^{qua}$	$\ln x_{fpdt}^{uv}$	$\ln x_{fpdt}^{val}$	$\ln x_{fpdt}^{qua}$	$\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0039)	−0.0013 (0.0041)	−0.0004 (0.0013)	−0.0001 (0.0010)	−0.00003 (0.0011)	−0.0001 (0.0003)
Asymptotic p	[0.566]	[0.750]	[0.748]	[0.924]	[0.974]	[0.857]
Bootstrap p	[0.686]	[0.851]	[0.740]	[0.931]	[0.978]	[0.860]
$EP_{dt} \times b_p$	−0.0044* (0.0022)	−0.0048* (0.0029)	0.0001 (0.0010)	−0.0009 (0.0006)	−0.0006 (0.0007)	−0.0001 (0.0004)
Asymptotic p	[0.052]	[0.096]	[0.903]	[0.153]	[0.409]	[0.697]
Bootstrap p	[0.185]	[0.391]	[0.937]	[0.177]	[0.443]	[0.734]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,313,047	21,313,037	21,519,511	21,313,047	21,313,037
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0019 (0.0037)	−0.0015 (0.0034)	−0.0004 (0.0019)	−0.0001 (0.0010)	−0.0001 (0.0010)	−0.0001 (0.0005)
Asymptotic p	[0.613]	[0.660]	[0.819]	[0.904]	[0.917]	[0.776]
Bootstrap p	[0.673]	[0.755]	[0.843]	[0.920]	[0.933]	[0.852]
$EP_{dt} \times b_p$	−0.0056** (0.0027)	−0.0070** (0.0030)	0.0004 (0.0011)	−0.0014** (0.0007)	−0.0014* (0.0008)	−0.00004 (0.0003)
Asymptotic p	[0.039]	[0.020]	[0.676]	[0.048]	[0.080]	[0.903]
Bootstrap p	[0.049]	[0.035]	[0.765]	[0.069]	[0.100]	[0.915]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,517,666	21,311,206	21,311,196	21,517,666	21,311,206	21,311,196
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963

Notes: Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv} = \ln x_{fpdt}^{val} - \ln x_{fpdt}^{qua}$). Quantity and unit-value regressions are estimated on the subset of observations with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

5.3 The collapsed panel, and what survives robust inference

Table 8 repeats the exercise on HS6–destination–year cells weighted by cell size. Three things change at once and it is worth naming them before reading the numbers. The unit of observation is a product–destination–year cell rather than a firm–product–destination–year shipment; the dependent variable is therefore a cell mean of log exports rather than an individual flow; and the fixed effects become $\theta_{pd} + \theta_{dt} + \theta_{pt}$, since firm identifiers no longer exist in the collapsed data. The last of these is the substantive change. The firm dimension of the absorbers is gone, so a shift in *which* firms ship dirty goods to a destination now enters the coefficient, where in Table 7 it was absorbed. The collapsed panel is not a robustness check on the full panel: it estimates a wider object, and the difference between the two is informative in its own right.¹⁷

On green the two panels agree, as they have agreed everywhere so far: -0.0046 (bootstrap $p = 0.65$) with the World Bank index, $+0.0018$ ($p = 0.39$) with TREND, and again nothing on quantity or unit value. On dirty they do not. The World Bank interaction is -0.0119 against -0.0044 in the full panel, a factor of 2.7, with an asymptotic p -value below 0.001. Since the two specifications differ by exactly the between-firm channel, the gap measures it: $(0.0119 - 0.0044)/0.0119 = 0.63$, so roughly three-fifths of the aggregate dirty association is a change in the composition of exporters rather than reallocation inside a given firm. This is the quantity that reconciles the present result with Zhu and C. Sun (2026). Their design keeps that channel; equation (1) removes it. Neither is wrong, and the two sets of results are not in conflict once the estimands are stated.

The collapsed panel is also where the design can be attacked with a third test. The wild cluster bootstrap already raises the dirty p -value from below 0.001 to 0.072, which is a large correction, but the bootstrap still takes the assignment of treatment as given and resamples only the residuals. The permutation test asks a blunter question. The environmental profile of each agreement — its depth and the years in which it takes effect — is detached from its destination and reassigned at random among the 23 treated destinations; the model is re-estimated; and the exercise is repeated a thousand times. The reported p -value is the share of these reassignments that produce a coefficient at least as large in absolute value as the observed one. The placebo distribution is not centred at zero (its mean is $+0.003$ for the dirty interaction), so the two-sided figure adds a right tail the alternative does not concern; the one-sided share of reassignments at least as negative as the observed coefficient is 0.08, and 0.17 after centring on the placebo mean. It is computationally expensive, which is why

¹⁷The two are related by an exact algebraic identity when the fixed effects are held the same. Estimating the firm-level panel under $pd + dt + pt$ reproduces the weighted collapsed-panel coefficient to nine significant digits, which is what makes the comparison in this section a comparison of fixed-effect structures rather than of datasets (Appendix A).

it is run on export value only, and it is discrete, because thirteen distinct profiles do not generate a smooth distribution. But it answers the question a reader of this table actually has: would this number have appeared if we had simply relabelled which of these destinations signed environmental chapters?

For the dirty coefficient the answer is that it would, 27.8 percent of the time, or 8.1 percent counting only reassignments as negative as the estimate. For green it is 59.7 percent. Under the DESTA control the two figures are 14.6 and 48.0 percent. This settles the question Section 5.2 could not, because the permutation distribution is built by reassigning whole agreement profiles rather than by partialling out a control, and it returns the same verdict under both depth measures (0.28 and 0.15). Appendix Table 18 makes the point from the other direction, assembling the dirty coefficient across all four sample–depth variants under a single inference method: the point estimate is uniformly negative, and the bootstrap p -value tracks the control variable rather than the data.

A leave-one-out exercise closes the argument, and it does not close it the way the phrase “driven by one country” usually implies. Table 9 re-estimates the dirty interaction dropping each treated destination in turn. The point estimate is remarkably stable: it sits between -0.0097 and -0.0133 in all 23 samples, and dropping large partners such as India or Pakistan moves it by one to four percent. What is fragile is the precision. Dropping Australia moves the coefficient by 13 percent, to -0.0103 , while nearly tripling its standard error from 0.0030 to 0.0087; dropping Korea doubles the standard error and gives -0.0097 . Australia and Korea are not outliers pulling the estimate — they supply the variation that identifies it — and both entered into force on 20 December 2015, so the precision of the dirty estimate rests on two destinations observed for eleven days of treatment, and without them the design cannot determine much of anything. Which of the two is pivotal depends, once again, on the depth control: under DESTA it is Korea rather than Australia. A coefficient whose decisive observation is selected by a modelling choice unrelated to environmental content is behaving the way the permutation test already suggested it was.

The same exercise on the green margin (Table 10) is quieter but not featureless. Dropping Switzerland moves the green coefficient to -0.011 (asymptotic $p < 0.001$), the one leave-one-out sample in which it reaches significance; dropping Korea or Australia moves it to $+0.003$ and $+0.002$, with standard errors roughly 1.5 and 1.9 times the baseline. The point estimate stays inside a narrow band for every other destination, but its precision, like the dirty coefficient’s, is not uniform across the sample.

One ambiguity we do not resolve. Australia and Korea are also the destinations with the deepest environmental coverage in the sample, so a dose–response reading — the effect is real and concentrated where the dose is largest — is not ruled out by the leave-one-out geometry

alone. Taken together with the permutation result, however, the reading we adopt is that the dirty association is present in the data but cannot be distinguished from what random relabelling of the same destinations produces.

A second consideration cuts the other way. The under-control argument of Section 4 — an imperfect depth control lets EP_{dt} absorb part of the effect of general agreement depth — biases both interactions upward. For the green margin this makes the null conservative; for the dirty margin it implies that the true coefficient is, if anything, more negative than the estimate. We note this rather than resolve it: the permutation test, which reassigns environmental and non-environmental content together, is unaffected by the quality of the control, and it is the test on which the reading above rests.

Table 8: Collapsed panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0046 (0.0069)	−0.0055 (0.0069)	0.0005 (0.0028)	0.0018 (0.0018)	0.0019 (0.0020)	−0.0001 (0.0007)
Asymptotic p	[0.512]	[0.426]	[0.854]	[0.320]	[0.360]	[0.868]
Bootstrap p	[0.649]	[0.525]	[0.851]	[0.390]	[0.433]	[0.867]
Permutation p	[0.597]			[0.160]		
$EP_{dt} \times b_p$	−0.0119*** (0.0029)	−0.0115* (0.0067)	0.0006 (0.0042)	0.0004 (0.0016)	−0.0004 (0.0019)	0.0009 (0.0005)
Asymptotic p	[0.000]	[0.087]	[0.880]	[0.826]	[0.837]	[0.076]
Bootstrap p	[0.072]	[0.250]	[0.947]	[0.857]	[0.886]	[0.169]
Permutation p	[0.278]			[0.817]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,672,709	3,672,709	3,681,023	3,672,709	3,672,709
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0043 (0.0043)	−0.0084* (0.0044)	0.0038 (0.0042)	0.0016 (0.0020)	0.0010 (0.0022)	0.0006 (0.0008)
Asymptotic p	[0.320]	[0.058]	[0.368]	[0.428]	[0.649]	[0.495]
Bootstrap p	[0.546]	[0.268]	[0.446]	[0.508]	[0.712]	[0.542]
Permutation p	[0.480]			[0.316]		
$EP_{dt} \times b_p$	−0.0113*** (0.0022)	−0.0097** (0.0045)	−0.0012 (0.0029)	−0.0002 (0.0014)	−0.0008 (0.0014)	0.0007 (0.0006)
Asymptotic p	[0.000]	[0.033]	[0.689]	[0.858]	[0.603]	[0.252]
Bootstrap p	[0.047]	[0.311]	[0.757]	[0.905]	[0.876]	[0.332]
Permutation p	[0.146]			[0.904]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,669,025	3,669,025	3,677,333	3,669,025	3,669,025
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966

Notes: Collapsed panel (product×destination×period cells, analytic weights), excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. Permutation p -values (treated-only design, $R = 1,000$) reported for export value only; columns (2)–(3) and (5)–(6) not permuted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 9: Leave-one-out: EP×dirty coefficient stability across 23 treated destinations

Excluded destination	WB depth control		DESTA depth control	
	Excl. HK/Macao (1)	Incl. HK/Macao (2)	Excl. HK/Macao (3)	Incl. HK/Macao (4)
<i>None (baseline)</i>	−0.0119***	−0.0189***	−0.0113***	−0.0082*
<i>None, extended dirty</i>	−0.0121***	−0.0191***	−0.0107***	−0.0071
<i>Excluding the three deepest (Peru, Switzerland, Korea)</i>	−0.0271**	−0.0456***	−0.0161	−0.0164
Bangladesh	−0.0118***	−0.0188***	−0.0113***	−0.0082*
Brunei	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Myanmar	−0.0118***	−0.0188***	−0.0112***	−0.0081*
Cambodia	−0.0119***	−0.0189***	−0.0112***	−0.0081*
India	−0.0120***	−0.0190***	−0.0115***	−0.0084**
Indonesia	−0.0119***	−0.0190***	−0.0119***	−0.0087**
Laos	−0.0119***	−0.0189***	−0.0113***	−0.0082*
Malaysia	−0.0124***	−0.0195***	−0.0110***	−0.0076*
Pakistan	−0.0114***	−0.0183***	−0.0106***	−0.0056
Philippines	−0.0118***	−0.0189***	−0.0115***	−0.0083*
Singapore	−0.0098**	−0.0166***	−0.0128***	−0.0100**
South Korea	−0.0097*	−0.0225**	−0.0104	−0.0085
Sri Lanka	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Thailand	−0.0120***	−0.0189***	−0.0115***	−0.0083*
Vietnam	−0.0122***	−0.0192***	−0.0112***	−0.0080*
Timor-Leste	−0.0119***	−0.0189***		
Iceland	−0.0119***	−0.0189***	−0.0112***	−0.0081*
Switzerland	−0.0133***	−0.0209***	−0.0115***	−0.0084*
Chile	−0.0121***	−0.0191***	−0.0104***	−0.0071
Costa Rica	−0.0114***	−0.0185***	−0.0109***	−0.0076*
Peru	−0.0133***	−0.0204***	−0.0121***	−0.0092**
Australia	−0.0103	−0.0245**	−0.0110***	−0.0048
New Zealand	−0.0119***	−0.0190***	−0.0113***	−0.0081*
Hong Kong		−0.0129***		−0.0112***
Macao		−0.0183***		−0.0082*
Observations	3,681,023	3,769,187	3,677,333	3,759,733
Clusters (<i>d</i>)	228	230	228	230

Notes: OLS on the collapsed panel. Dependent variable: log total exports from China. The coefficient shown is $\hat{\beta}$ on EP×dirty, re-estimated dropping one treated destination at a time. FE: $pd + dt + pt$. Standard errors (not shown) clustered by destination. Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. Columns (1)–(2) use WB total depth as the control; columns (3)–(4) use DESTA depth. Observations and clusters refer to the baseline; individual rows differ slightly due to the dropped destination. Excluding the three deepest: estimated dropping Peru, Switzerland and Korea jointly. Timor-Leste rows blank in columns (3)–(4) (absent from DESTA). Hong Kong and Macao rows blank in columns (1) and (3) (already excluded). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 10: Leave-one-out: EP×green coefficient stability across 23 treated destinations

Excluded destination	EP×green
<i>None (baseline)</i>	−0.0046
<i>None, extended dirty</i>	−0.0047
<i>Excluding the three deepest (Peru, Switzerland, Korea)</i>	−0.0255**
Bangladesh	−0.0045
Brunei	−0.0046
Myanmar	−0.0046
Cambodia	−0.0046
India	−0.0045
Indonesia	−0.0046
Laos	−0.0046
Malaysia	−0.0048
Pakistan	−0.0045
Philippines	−0.0042
Singapore	+0.0031
South Korea	+0.0031
Sri Lanka	−0.0045
Thailand	−0.0047
Vietnam	−0.0047
Timor-Leste	−0.0046
Iceland	−0.0046
Switzerland	−0.0111***
Chile	−0.0045
Costa Rica	−0.0038
Peru	−0.0058
Australia	+0.0017
New Zealand	−0.0034
Observations	3,681,023
Clusters (d)	228

Notes: OLS on the collapsed panel, WB total depth control, excluding Hong Kong and Macao (same sample as column (1) of Table 9). The coefficient shown is $\hat{\beta}$ on EP×green, re-estimated dropping one treated destination at a time. FE: $pd + dt + pt$. Standard errors (not shown) clustered by destination. Excluding the three deepest: estimated dropping Peru, Switzerland and Korea jointly. Dropping Switzerland moves the coefficient to −0.011 (asymptotic $p < 0.001$); dropping Korea or Australia moves it to +0.003 and +0.002, with standard errors roughly 1.5 and 1.9 times the baseline. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

5.4 Changing the comparison group

Everything so far compares green and dirty products against neutral products, and agreement destinations against non-agreement destinations. Both comparisons can be questioned, and Table 11 replaces them one at a time. Every cell in it is a separate estimation and every p -value is a wild cluster bootstrap, so the forty cells are directly comparable; the full six-column versions of the three subsamples are in Appendices Q.1, Q.2 and Q.3.

Neutral products may be poor counterfactuals for green ones if the two sit in different regions of the product space. *C-prod-HS4* therefore restricts the neutral group to products in the same HS4 heading as a green code, so that a solar panel is compared with its four-digit neighbours rather than with the entire neutral universe. *CEM* addresses the symmetric concern about destinations: China does not sign agreements at random, so partners are matched to non-partners on log GDP per capita, growth and the MFN tariff in 2000, leaving 57 destination clusters comparable on pre-treatment observables. *Deep versus shallow* removes the untreated comparison group altogether, restricting the sample to the 23 PTA partners so that a shallow environmental chapter becomes the counterfactual for a deep one.

The two margins behave differently across the three, and the difference is the point of the table. Across the twelve green cells of the three subsamples the coefficient stays between -0.0023 and $+0.0005$ and its bootstrap p -value never falls below 0.58; including the two reference rows, the range widens only to $[-0.0046, +0.0018]$ and the lowest p -value is 0.39. An artifact of the comparison group would move when the comparison group is replaced three times over; this does not move.

The dirty coefficient survives none of the three replacements. Under matching it is -0.0041 ($p = 0.25$) and under the partners-only design -0.0034 ($p = 0.50$), so it keeps its sign but loses whatever standing it had. Under the within-heading product comparison it changes sign altogether, to $+0.0066$. Narrowing the product comparison does not shrink the dirty estimate toward zero so much as scramble it, which is what one expects of a coefficient that is not identifying a stable object. Together with the permutation result of Section 5.3, the reading that fits the evidence is that the dirty coefficient is a feature of the specification in which it appears rather than a composition effect that would surface wherever one looked.

Further variations — adding product–destination–year controls for tariffs, market concentration and antidumping exposure, dropping the eleven destinations covered by the ASEAN cohort, re-including Hong Kong and Macao, restricting to products with common support, and replacing the OECD list of environmental goods with the narrower APEC list — are reported in Appendices K, O and J. Appendix P replaces the binary dirty indicator with a continuous measure of product-level CO₂ emission intensity taken from Shapiro (2021). None moves the green coefficient outside the band already described, and the continuous

dirty measure fades under the bootstrap exactly as the binary one does.

Table 11: Control groups and estimation designs: EP interactions across subsamples

Sample / design	WB EP depth		TREND EP count		Obs.	Clusters
	$\times g_p$	$\times b_p$	$\times g_p$	$\times b_p$		
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
Full panel (reference)	−0.0023	−0.0044	−0.0001	−0.0009	21,519,511	225
bootstrap p	[0.686]	[0.185]	[0.931]	[0.177]		
Collapsed panel	−0.0046	−0.0119	+0.0018	+0.0004	3,681,023	228
bootstrap p	[0.649]	[0.072]	[0.390]	[0.857]		
C-prod-HS4	−0.0009	+0.0066	+0.0005	+0.0024	3,772,855	215
bootstrap p	[0.860]	[0.435]	[0.730]	[0.734]		
CEM-matched destinations	−0.0023	−0.0041	−0.0006	−0.0011	13,992,396	57
bootstrap p	[0.610]	[0.251]	[0.609]	[0.204]		
Deep vs. shallow (partners only)	−0.0022	−0.0034	−0.0004	−0.0006	5,262,293	23
bootstrap p	[0.585]	[0.499]	[0.751]	[0.739]		
<i>Panel B — DESTA depth control</i>						
Full panel (reference)	−0.0019	−0.0056	−0.0001	−0.0014	21,517,666	225
bootstrap p	[0.673]	[0.049]	[0.920]	[0.069]		
Collapsed panel	−0.0043	−0.0113	+0.0016	−0.0002	3,677,333	228
bootstrap p	[0.546]	[0.047]	[0.508]	[0.905]		
C-prod-HS4	−0.0009	+0.0033	+0.0004	+0.0015	3,772,321	214
bootstrap p	[0.855]	[0.850]	[0.817]	[0.735]		
CEM-matched destinations	−0.0016	−0.0043	−0.0005	−0.0013	13,992,396	57
bootstrap p	[0.676]	[0.121]	[0.676]	[0.125]		
Deep vs. shallow (partners only)	−0.0016	−0.0042	−0.0003	−0.0009	5,260,451	23
bootstrap p	[0.616]	[0.173]	[0.784]	[0.386]		

Notes: Dependent variable: log export value ($\ln x_{fpdt}^{val}$). Every cell is a separate estimation of equation (1) on the indicated sample. Fixed effects: $\theta_{fpd} + \theta_{fdt} + \theta_{pt}$ (firm-level samples), $\theta_{pd} + \theta_{dt} + \theta_{pt}$ (collapsed panel). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Hong Kong and Macao excluded. Wild cluster bootstrap ($B = 9,999$) p -values in brackets, clustered by destination. Asymptotic p -values in the full tables (Appendix Tables 34, 35 and 36).

C-prod-HS4: neutral products restricted to the 106 HS4 headings containing at least one green code. *CEM:* destinations matched on log GDP per capita, growth and MFN tariff in 2000. *Deep vs. shallow:* PTA partner destinations only; 23 clusters, of which 7 shallow.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database; DESTA.

5.5 What is inside the agreements

One reading of the null so far is that it is an artifact of aggregation. The indices used to this point add up every environmental provision in an agreement regardless of what it does, and most of what they add up is cooperation language, institutional machinery and general environmental statements — text with no channel through which it could move a product

mix. Only a small minority of provisions carry a trade mechanism: tariff reductions on environmental goods, non-regression clauses that prevent a partner from relaxing standards to attract trade, binding market-access commitments. If the few clauses that could bite are diluted by the many that cannot, a null on the aggregate index would say nothing about whether environmental content matters.

The dilution is severe, and Table 12 quantifies it. The World Bank index records 150 provisions across the treated destinations, and 8 of them — 5.3 percent — fall in sub-indices with a trade mechanism. TREND, which codes at finer granularity, gives 4 of 437, or 0.9 percent. At the level of the agreement, a binding environmental obligation appears in exactly one of the fourteen Chinese PTAs in the database, the 2015 agreement with Korea. Appendix Table 25 sets out what each sub-index contains; the sub-indices themselves are constructed here from the raw provision-level codings and are not variables native to either database.

Table 12: Trade-mechanism content as a share of total environmental provisions

	Checked	Trade-mechanism	Share
<i>Panel A: Provision-level</i>			
WB: Green Liberalization + Standards Non-Regression	150	8	5.3%
TREND: Green Market Access	437	4	0.9%
	Agreements	With clause	Share
<i>Panel B: Agreement-level</i>			
TREND: Binding environmental obligations	14	1	7.1%

Notes: Panel A sums each depth index across the 25 PTA destinations (including Hong Kong and Macao) at each destination’s maximum value. “Checked” is the total number of provisions present; “Trade-mechanism” counts those in sub-indices with a direct channel to trade flows. Panel B counts Chinese PTAs in the TREND database; binding obligations are the TREND variable X5.01.01 **Binding obligations**, coded 1 when the environmental chapter states an obligation in binding terms, and the single agreement carrying it is the China–Korea PTA (2015). The two WB trade-mechanism sub-indices are non-zero in the same three destination–years (Korea 2015, Switzerland 2014–15), so the 8 counts one collinear pair twice. Source: World Bank DTA Database; TREND Database.

The structural consequence is stark. The two World Bank sub-indices with a trade mechanism are non-zero in only three destination–years, always in the same proportion, which

makes them perfectly collinear ($\rho = 1.000$) and impossible to identify separately. Brandi et al. (2020) report a maximum variance inflation factor of 4.6 across provision types on a panel of 680 agreements; with thirteen distinct profiles there is no equivalent here. What can still be done is to replace the aggregate index with one sub-index at a time and re-estimate. Table 13 does this on the full panel for seven sub-indices, two from the World Bank coding and five from TREND. The coefficients are not partial effects holding other clause types fixed; each absorbs whatever correlated variation moves with it, so the table is a set of separate comparisons rather than a decomposition. It should be read for pattern, not for magnitude.

The pattern is the one already familiar. On the green margin nothing registers: the two trade-mechanism sub-indices give +0.0085 (World Bank Green Liberalization, $p = 0.86$) and +0.0158 (TREND Green Market Access, $p = 0.38$), and the remaining five are equally silent. On the dirty margin two sub-indices are marginally negative — enforcement and dispute settlement attached to the environmental chapter, at -0.0243 ($p = 0.040$) in the World Bank coding and -0.0107 ($p = 0.081$) in TREND. This is the same asymptotics-only signal documented in Section 5.2, now on a smaller and more collinear regressor, and it should be discounted accordingly: the wild cluster bootstrap has not been run on the full-panel sub-indices, so these two figures rest on precisely the inference this paper argues against.¹⁸

The conclusion of this section is a negative one, and it is the honest one: this sample cannot tell whether the null reflects the absence of an effect or the absence of provisions capable of producing one. What it can establish is that the two are observationally equivalent here, and that a null on the aggregate index is exactly what Brandi et al. (2020)’s own heterogeneity results would predict for a set of agreements composed almost entirely of cooperation clauses. Their effects come from the subset of agreements with explicitly liberalizing or restrictive clauses, identified separately because 680 agreements supply enough cross-agreement variation in provision type. The Chinese agreements of this period, on the same coding, do not.

¹⁸On the collapsed panel, where the bootstrap is available for the sub-indices (Appendix Table 24), the only coefficient that survives it is the TREND regulatory-space index, at +0.024 on green ($p = 0.045$) and +0.023 on dirty ($p = 0.021$). Both margins move together and by the same amount, so the two interactions are statistically indistinguishable: this is not a shift in composition toward cleaner goods but a proportional rise in both categories against the neutral baseline. The regulatory-space index also accounts for 36 percent of the TREND count across treated destination-years and correlates 0.87 with the aggregate TREND count and 0.89 with World Bank non-environmental depth within destinations and years (0.70 in the raw data), which makes it a poor placebo rather than a clean channel. In a battery of roughly forty tests, two coefficients significant at the five percent level are what chance alone delivers.

Table 13: Full panel — sub-index decomposition

	WB sub-indices		TREND sub-indices				
	(1) Green Lib.	(2) Enf. DSM	(3) Green Mkt.	(4) Enf. DSM	(5) Hard	(6) Soft	(7) Reg. Space
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>							
$EP_{\text{sub}} \times g_p$	0.0085 (0.0467)	−0.0028 (0.0208)	0.0158 (0.0178)	0.0020 (0.0083)	0.0001 (0.0019)	−0.0037 (0.0085)	0.0049 (0.0095)
$EP_{\text{sub}} \times b_p$	−0.0019 (0.0335)	−0.0243** (0.0117)	−0.0088 (0.0153)	−0.0107* (0.0061)	−0.0017 (0.0012)	−0.0047 (0.0070)	0.0072 (0.0075)
θ_{fpd}	✓	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511
Clusters (d)	225	225	225	225	225	225	225
Adj. R^2	0.744	0.744	0.744	0.744	0.744	0.744	0.744
<i>Panel B — DESTA depth control</i>							
$EP_{\text{sub}} \times g_p$	0.0114 (0.0431)	−0.0002 (0.0200)	0.0163 (0.0159)	0.0025 (0.0086)	0.0003 (0.0019)	−0.0032 (0.0081)	0.0054 (0.0090)
$EP_{\text{sub}} \times b_p$	−0.0193** (0.0309)	−0.0389** (0.0166)	−0.0166 (0.0138)	−0.0135** (0.0064)	−0.0027* (0.0014)	−0.0086** (0.0042)	−0.0060 (0.0075)
θ_{fpd}	✓	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	21,283,431	21,283,431	21,283,431	21,283,431	21,283,431	21,283,431	21,283,431
Clusters (d)	225	225	225	225	225	225	225
Adj. R^2	0.745	0.745	0.745	0.745	0.745	0.745	0.745

Notes: Full firm-level panel (excluding Hong Kong and Macao); sub-indices are set to zero for destinations without an agreement. Panel B drops observations for which DESTA depth is missing while WB depth is positive (Timor-Leste, not covered by DESTA), the same restriction applied elsewhere in the paper, which accounts for the 236,080-observation difference from Panel A. Each column estimates a separate model replacing the aggregate EP index with one sub-index; the non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. Fixed effects: firm-product-destination (θ_{fpd}), firm-destination-year (θ_{fdt}), product-year (θ_{pt}). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Adjusted R^2 ; the unadjusted R^2 of the reference specification is 0.873 (Table 7).

WB sub-indices: (1) Green Liberalization; (2) Enforcement & DSM. TREND sub-indices: (3) Green Market Access; (4) Enforcement & DSM; (5) Hard obligations; (6) Soft obligations; (7) Regulatory Space.

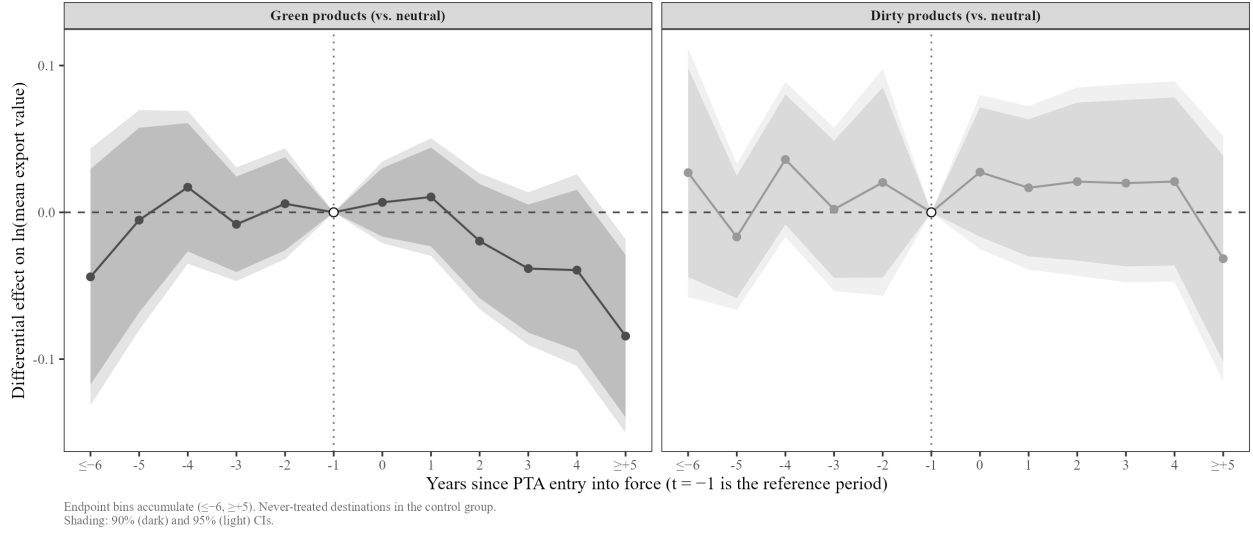
5.6 Dynamics

Figure 4 plots the green and dirty gap relative to neutral products around entry into force, with never-treated destinations as controls. No pre-treatment coefficient is individually distinguishable from zero on either margin; the green leads rise from -0.04 at $t \leq -6$ to zero at $t = -1$, and the $t \leq -6$ bin is identified by the 2005 and later cohorts only, since the 2002 cohort has two pre-entry years, which is the standard falsification of the identifying assumption. It should be read as exactly that and no more: parallel trends is a statement about the counterfactual path after treatment, and a flat pre-trend rules out the most detectable violation without establishing that the assumption holds thereafter.

A mild negative drift appears for green products in the late post-period, but it is identified only by the cohorts that entered by 2010, and dominated by the two early ones (Bangkok 2002 and ASEAN 2005) — and it disappears under the L. Sun and Abraham (2021) interaction-weighted estimator, whose aggregated treatment effect is -0.042 ($p = 0.27$) on the green gap and $+0.073$ ($p = 0.28$) on the dirty gap. Both event studies answer a narrower question than equation (1): they binarise treatment and discard depth; the Sun–Abraham version additionally operates on a gap collapsed across products, so it is a diagnostic for staggered timing rather than a replication.

A second concern is that the fixed effects do not absorb a destination-specific shock to green products that grows over time — a country signing environmental clauses precisely as its demand for green goods is rising. Allowing each destination its own linear trend interacted with product type over the full sample period leaves the World Bank results unchanged and turns the TREND green interaction significantly negative, the only green coefficient among the robustness checks of Appendix H to survive the bootstrap. Estimating destination-specific trends on pre-treatment years only, projecting them forward and re-estimating on the detrended outcome leaves no coefficient significant under the bootstrap. The detrended dirty interaction, however, changes sign and grows to $+0.06$ per provision (bootstrap $p = 0.61$), which says that the pre-entry slopes extrapolated from two to four years are too imprecise to detrend anything: this variant bounds nothing and is reported for completeness. Appendix F tabulates the event study, Appendix G reports the interaction-weighted estimator in full, and Appendix H the two trend specifications.

Figure 4: Event study: green and dirty vs. neutral products around PTA entry into force



Notes: Collapsed panel; FE $pd + dt + pt$; binary treatment, no depth control; destination-clustered 95% confidence bands; $t = -1$ is the reference period; endpoint bins accumulate. Coefficients are tabulated in Appendix Table 21. Source: Chinese Customs Office; World Bank DTA Database.

5.7 The extensive margin

Everything above conditions on positive flows, so an agreement that created green exports to a market previously served by none would leave no trace in it. Table 14 estimates the same interactions by Poisson pseudo-maximum likelihood on an HS6–destination–year grid completed with zeros (Santos Silva and Tenreyro, 2006), of which 7.9 million cells out of 8.2 survive the estimator’s removal of separated and singleton observations.

The green interaction is $+0.0015$ ($p = 0.74$) with the World Bank index and $+0.0001$ ($p = 0.95$) with TREND. There is no trade creation to detect. The dirty interaction is -0.030 ($p = 0.057$) and $+0.003$ ($p = 0.55$), marginal at best and, once again, only under one of the two codings — the same pattern the intensive margin produced, which is at least internally consistent.

One limitation deserves stating. This grid detects a product newly exported to a destination, not a new firm beginning to export a product that other Chinese firms already ship there, which is a common form of firm-level entry. The extensive margin is probed, not exhausted.

Table 14: PPML — extensive margin (collapsed panel, zero-filled)

	WB EP depth (EP_{dt}^{WB})		TREND EP count (EP_{dt}^{TR})	
	(1)	(2)	(3)	(4)
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>				
$EP_{dt} \times g_p$	0.0015 (0.0046)	0.0029 (0.0061)	0.0001 (0.0019)	−0.0001 (0.0021)
Asymptotic p	[0.738]	[0.630]	[0.951]	[0.972]
$EP_{dt} \times b_p$	−0.0301* (0.0158)	−0.0268 (0.0178)	0.0028 (0.0047)	0.0048 (0.0050)
Asymptotic p	[0.057]	[0.133]	[0.550]	[0.335]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,895,543	8,030,137	7,895,543	8,030,137
Clusters (d)	225	228	225	228
<i>Panel B — DESTA depth control</i>				
$EP_{dt} \times g_p$	0.0053 (0.0045)	0.0064 (0.0050)	0.0010 (0.0013)	0.0008 (0.0015)
Asymptotic p	[0.232]	[0.205]	[0.442]	[0.565]
$EP_{dt} \times b_p$	−0.0186 (0.0169)	−0.0108 (0.0198)	0.0027 (0.0053)	0.0046 (0.0050)
Asymptotic p	[0.270]	[0.585]	[0.612]	[0.358]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,878,784	8,003,772	7,878,784	8,003,772
Clusters (d)	225	228	225	228

Notes: PPML on the HS6–destination–year grid completed with zeros (). Fixed effects: θ_{pd} , θ_{dt} , θ_{pt} . Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

6 Conclusion

China’s trade agreements did not shift the composition of its exports between 2000 and 2015. On the green margin, the finding is a *bounded null*: the coefficient is indistinguishable from zero in both panels, under both provision codings, under both depth controls, on value, quantity and unit value alike, and under three separate replacements of the comparison group. It is robust to bootstrap and permutation inference, and it is bounded from above by roughly one quarter of the reference estimate in the aggregate literature, on an indicative per-provision comparison. This is not a failure to reject. It is a circumscribed statement about the absence of a specific effect — the kind the literature has documented elsewhere under different conditions.

On the dirty margin the finding is weaker and we state it as such. The coefficient is negative almost everywhere and its magnitude is stable, but its statistical standing is not. Under the World Bank measure of general agreement depth the wild cluster bootstrap takes it well clear of conventional thresholds; under the DESTA measure it does not, because the two controls leave different amounts of independent variation in environmental depth. Two further pieces of evidence resolve the ambiguity in the same direction. The permutation test, which reassigns whole agreement profiles and therefore does not depend on how well the depth control is measured, returns p -values between 0.15 and 0.38 in every variant. And the leave-one-out exercise shows that the point estimate is stable across all 23 exclusions (-0.0097 to -0.0133) while the standard error doubles or triples when one of two destinations is dropped — and which two depends, again, on the depth control. Precision, not the estimate, is what those destinations supply. We do not read this as an identified composition effect, and we do not claim it is definitively absent either: the honest summary is that the design cannot separate it from a placebo.

The central reading is about content. Environmental provisions affect trade when they are specific, binding, and targeted at a product category — Abman, Lundberg, and Ruta (2024) show this for deforestation, Brandi et al. (2020) for trade composition. China’s chapters of this period were thin and cooperation-heavy: the two WB sub-indices with a direct trade mechanism are perfectly collinear in-sample and appear in only three treated country-years. The policy implication is direct: a chapter in an agreement is not itself the instrument. What the chapter contains is. This reading is an inference across studies, not a causal finding of the present design: the current specification cannot separate the effect of provision content from that of agreement selection within the Chinese PTA context.

Methodologically, the paper adds a cautionary case. With treatment concentrated across roughly a dozen agreements, asymptotically significant composition effects can emerge and

dissolve under honest inference. Bootstrap, permutation, and leave-one-out — applied not as robustness appendices but as the primary mode of inference — should be standard practice for this class of design.

Three limitations should be noted. The preferential tariff rate was not obtainable. For the green margin its omission can only bias the estimate upward; for the dirty margin it is a genuine confounder, since the sectors we call dirty are those most often excluded from tariff liberalisation, and the negative dirty coefficient should be read with that in mind. A continuous-dose estimator in the spirit of Callaway, Goodman-Bacon, and Sant’Anna (2024) is not implemented; the TWFE coefficient is a weighted average of cohort-specific effects, though the permutation test is agnostic about this weighting. The extensive margin is probed at the product–destination level: new firm entry into a destination where other Chinese firms already export the same green product is not detected.

References

- Abadie, Alberto et al. (2023). “When should you adjust standard errors for clustering?” In: *Quarterly Journal of Economics* 138.1, pp. 1–35. DOI: 10.1093/qje/qjac038.
- Abman, Ryan, Clark Lundberg, and Michele Ruta (2024). “The effectiveness of environmental provisions in regional trade agreements”. In: *Journal of the European Economic Association* 22.6, pp. 2507–2548. DOI: 10.1093/jeea/jvae023.
- Antweiler, Werner, Brian R. Copeland, and M. Scott Taylor (2001). “Is free trade good for the environment?” In: *American Economic Review* 91.4, pp. 877–908. DOI: 10.1257/aer.91.4.877.
- Baier, Scott L. and Jeffrey H. Bergstrand (2007). “Do free trade agreements actually increase members’ international trade?” In: *Journal of International Economics* 71.1, pp. 72–95. DOI: 10.1016/j.jinteco.2006.02.005.
- (2009). “Estimating the effects of free trade agreements on international trade flows using matching econometrics”. In: *Journal of International Economics* 77.1, pp. 63–76. DOI: 10.1016/j.jinteco.2008.09.006.
- Bergé, Laurent (2018). “Efficient estimation of maximum likelihood models with multiple fixed-effects: the R package FENmlm”. In: *CREA Discussion Papers* 13. CRAN package `fixest`.
- Berger, Axel et al. (2020). “The trade effects of environmental provisions in preferential trade agreements”. In: *International Trade, Investment, and the Sustainable Development Goals*. Ed. by Cosimo Beverelli, Jürgen Kurtz, and Damian Raess. Cambridge University Press, pp. 111–139. DOI: 10.1017/9781108881364.006.

- Brandi, Clara et al. (2020). “Do environmental provisions in trade agreements make exports from developing countries greener?” In: *World Development* 129, p. 104899. DOI: 10.1016/j.worlddev.2020.104899.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant’Anna (2024). *Difference-in-differences with a continuous treatment*. Working Paper 32117. National Bureau of Economic Research. DOI: 10.3386/w32117.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller (2008). “Bootstrap-based improvements for inference with clustered errors”. In: *Review of Economics and Statistics* 90.3, pp. 414–427. DOI: 10.1162/rest.90.3.414.
- Chaisemartin, Clément de and Xavier D’Haultfœuille (2020). “Two-way fixed effects estimators with heterogeneous treatment effects”. In: *American Economic Review* 110.9, pp. 2964–2996. DOI: 10.1257/aer.20181169.
- Cherniwchan, Jevan (2017). “Trade liberalization and the environment: Evidence from NAFTA and U.S. manufacturing”. In: *Journal of International Economics* 105, pp. 130–149. DOI: 10.1016/j.jinteco.2017.01.005.
- Copeland, Brian R., Joseph S. Shapiro, and M. Scott Taylor (2022). “Globalization and the environment”. In: *Handbook of International Economics*. Ed. by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff. Vol. 5. Elsevier, pp. 61–146. DOI: 10.1016/bs.hesint.2022.02.002.
- Copeland, Brian R. and M. Scott Taylor (1994). “North-South trade and the environment”. In: *Quarterly Journal of Economics* 109.3, pp. 755–787. DOI: 10.2307/2118421.
- Correia, Sergio (2017). *Linear models with high-dimensional fixed effects: An efficient and feasible estimator*. Working Paper.
- Correia, Sergio, Paulo Guimarães, and Tom Zylkin (2020). “Fast Poisson estimation with high-dimensional fixed effects”. In: *Stata Journal* 20.1, pp. 95–115. DOI: 10.1177/1536867X20909691.
- Dür, Andreas, Leonardo Baccini, and Manfred Elsig (2014). “The design of international trade agreements: Introducing a new dataset”. In: *Review of International Organizations* 9.3, pp. 353–375. DOI: 10.1007/s11558-013-9179-8.
- Eckel, Carsten and J. Peter Neary (2010). “Multi-product firms and flexible manufacturing in the global economy”. In: *Review of Economic Studies* 77.1, pp. 188–217. DOI: 10.1111/j.1467-937X.2009.00573.x.
- Feenstra, Robert C. and Gordon H. Hanson (2004). “Intermediaries in entrêpot trade: Hong Kong re-exports of Chinese goods”. In: *Journal of Economics & Management Strategy* 13.1, pp. 3–35. DOI: 10.1111/j.1430-9134.2004.00002.x.

- Felbermayr, Gabriel, Sonja Peterson, and Joschka Wanner (2025). “Trade and the environment, trade policies and environmental policies—How do they interact?” In: *Journal of Economic Surveys* 39.3, pp. 1148–1184. DOI: 10.1111/joes.12628.
- Frankel, Jeffrey (2009). *Environmental effects of international trade*. Faculty Research Working Paper RWP09-006. Harvard Kennedy School.
- Freund, Caroline L. and Emanuel Ornelas (2010). “Regional trade agreements”. In: *Annual Review of Economics* 2, pp. 139–166. DOI: 10.1146/annurev.economics.102308.124455.
- Goodman-Bacon, Andrew (2021). “Difference-in-differences with variation in treatment timing”. In: *Journal of Econometrics* 225.2, pp. 254–277. DOI: 10.1016/j.jeconom.2021.03.014.
- Gutsch, Michelle et al. (2025). “Effects of environmental provisions in international trade agreements on businesses and economies – a systematic review”. In: *Sustainability Accounting, Management and Policy Journal* 16.7, pp. 1–27. DOI: 10.1108/SAMPJ-02-2024-0122.
- Hofmann, Claudia, Alberto Osnago, and Michele Ruta (2017). *Horizontal depth: A new database on the content of preferential trade agreements*. Policy Research Working Paper 7981. World Bank. DOI: 10.1596/1813-9450-7981.
- Iacus, Stefano M., Gary King, and Giuseppe Porro (2012). “Causal inference without balance checking: Coarsened exact matching”. In: *Political Analysis* 20.1, pp. 1–24. DOI: 10.1093/pan/mpr013.
- Levinson, Arik and M. Scott Taylor (2008). “Unmasking the pollution haven effect”. In: *International Economic Review* 49.1, pp. 223–254. DOI: 10.1111/j.1468-2354.2008.00478.x.
- Low, Patrick and Alexander Yeats (1992). *Do “dirty” industries migrate?* Discussion Paper 159. World Bank, pp. 89–104.
- MacKinnon, James G. and Matthew D. Webb (2017). “Wild bootstrap inference for wildly different cluster sizes”. In: *Journal of Applied Econometrics* 32.2, pp. 233–254. DOI: 10.1002/jae.2508.
- Mani, Muthukumara and David Wheeler (1998). “In search of pollution havens? Dirty industry in the world economy, 1960 to 1995”. In: *Journal of Environment & Development* 7.3, pp. 215–247. DOI: 10.1177/107049659800700302.
- Mattoo, Aaditya, Alen Mulabdic, and Michele Ruta (2022). “Trade creation and trade diversion in deep agreements”. In: *Canadian Journal of Economics* 55.3, pp. 1598–1637. DOI: 10.1111/caje.12611.

- Mattoo, Aaditya, Nadia Rocha, and Michele Ruta, eds. (2020). *Handbook of Deep Trade Agreements*. Washington, DC: World Bank. DOI: 10.1596/978-1-4648-1539-3.
- Morin, Jean-Frédéric, Joost Pauwelyn, and James Hollway (2017). “The trade regime as a complex adaptive system: Exploration and exploitation of environmental norms in trade agreements”. In: *Journal of International Economic Law* 20.2, pp. 365–390. DOI: 10.1093/jiel/jgx013.
- Porter, Michael E. and Claas van der Linde (1995). “Toward a new conception of the environment-competitiveness relationship”. In: *Journal of Economic Perspectives* 9.4, pp. 97–118. DOI: 10.1257/jep.9.4.97.
- Roodman, David et al. (2019). “Fast and wild: Bootstrap inference in Stata using boottest”. In: *Stata Journal* 19.1, pp. 4–60. DOI: 10.1177/1536867X19830877.
- Santos Silva, João M. C. and Silvana Tenreyro (2006). “The log of gravity”. In: *Review of Economics and Statistics* 88.4, pp. 641–658. DOI: 10.1162/rest.88.4.641.
- Sauvage, Jehan (2014). *The stringency of environmental regulations and trade in environmental goods*. Trade and Environment Working Paper 2014/03. OECD. DOI: 10.1787/5jxrjn7xsnmq-en.
- Shapiro, Joseph S. (2021). “The environmental bias of trade policy”. In: *Quarterly Journal of Economics* 136.2, pp. 831–886. DOI: 10.1093/qje/qjaa042.
- Sun, Liyang and Sarah Abraham (2021). “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects”. In: *Journal of Econometrics* 225.2, pp. 175–199. DOI: 10.1016/j.jeconom.2020.09.006.
- Wolfers, Justin (2006). “Did unilateral divorce laws raise divorce rates? A reconciliation and new results”. In: *American Economic Review* 96.5, pp. 1802–1820. DOI: 10.1257/aer.96.5.1802.
- Young, Alwyn (2019). “Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results”. In: *Quarterly Journal of Economics* 134.2, pp. 557–598. DOI: 10.1093/qje/qjy029.
- Yue, Wen and Qingxia Lin (2024). “PTA environmental provisions and the export of energy-consuming products: Evidence from China”. In: *Environment, Development and Sustainability*. DOI: 10.1007/s10668-024-05852-3.
- Zhang, Qiang et al. (2017). “Transboundary health impacts of transported global air pollution and international trade”. In: *Nature* 543.7647, pp. 705–709. DOI: 10.1038/nature21712.
- Zhu, Yajun and Churen Sun (2025). “Environmental provisions in preferential trade agreements and export product quality of Chinese firms”. In: *Journal of Asian Economics* 101, p. 102048. DOI: 10.1016/j.asieco.2025.102048.

Zhu, Yajun and Churen Sun (2026). “Greener trade agreements and green transformation”.
In: *China & World Economy* 34.1, pp. 41–74. DOI: 10.1111/cwe.70005.

A Equivalence of the collapsed and firm-level panels

The collapsed panel aggregates firm–HS6–destination–year observations to HS6–destination–year cells with outcome equal to the within-cell mean of log exports and weights equal to cell counts. Under fixed effects $pd + dt + pt$ the weighted cell-level regression is algebraically identical to the unweighted micro-level regression under the same fixed effects. Verification to seven significant figures: the $EP \times \text{green}$ coefficient is -0.0045685 in both; the $EP \times \text{dirty}$ coefficient is -0.011873 in both. The result is exact up to floating-point precision and holds by construction of the collapse and weights, not by approximation. It is what licenses the reading of Section 5.3: the gap between Tables 7 and 8 is a difference of fixed-effect structures, not of datasets.

B Treated destinations

Table 15 lists the 25 destinations covered by a Chinese PTA, including Hong Kong and Macao, which the main sample excludes, their agreement entry years, and the maximum EP depth in the sample period for both the WB and TREND indices.

Table 15: Treated destinations: PTA entry-into-force year and environmental content

Destination	Dataset code	Year of entry into force	Maximum EP depth		
			WB	/	TREND
Bangladesh	103	2002	1	/	1
India	111	2002	1	/	1
South Korea	133	2002	17	/	72
Laos	119	2002	6	/	4
Sri Lanka	134	2002	1	/	1
Hong Kong	110	2003	4	/	5
Macao	121	2003	5	/	5
Brunei	105	2005	6	/	4
Cambodia	107	2005	6	/	4
Timor-Leste	144	2005	6	/	4
Indonesia	112	2005	6	/	4
Malaysia	122	2005	6	/	4
Myanmar	106	2005	6	/	4
Philippines	129	2005	6	/	4
Singapore	132	2005	7	/	15
Thailand	136	2005	6	/	4
Vietnam	141	2005	6	/	4
Chile	412	2006	5	/	30
Pakistan	127	2007	3	/	12
New Zealand	609	2008	7	/	53
Peru	434	2010	12	/	47
Costa Rica	415	2011	4	/	50
Iceland	322	2014	6	/	28
Switzerland	331	2014	14	/	53
Australia	601	2015	3	/	24
Total	25 destinations				

Notes: Each row is a destination covered by a Chinese PTA in force between 2000 and 2015. EP depth = count of environmental provisions according to two independent codings: World Bank (WB) and TREND. The 11 ASEAN destinations share the same agreement (2005) and therefore the same EP values; the truly distinct agreements are 14. Hong Kong and Macao excluded from the main sample due to their entrepôt function.

Source: World Bank DTA Database; TREND Database.

C Inference: full bootstrap and permutation tables

Table 16 reports the wild cluster bootstrap p -values for the four headline coefficients across both panels, both indices, and all four sample-depth variants; Table 17 does the same for the permutation test on export value. Both are the complete versions of the figures quoted in Sections 5.2 and 5.3.

Table 16: Wild cluster bootstrap p -values (9,999 replications)

	(1)	(2)	(3)	(4)
<i>Panel A — collapsed panel (WB index)</i>				
EP depth $\times$ Green: coefficient	-0.00457	0.00173	-0.00433	-0.00675
Bootstrap p -value	0.649	0.825	0.546	0.484
EP depth $\times$ Dirty: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p -value	0.072	0.005	0.047	0.195
<i>Panel B — collapsed panel (TREND index)</i>				
EP count $\times$ Green: coefficient	0.00181	0.00178	0.00161	0.00017
Bootstrap p -value	0.390	0.414	0.508	0.949
EP count $\times$ Dirty: coefficient	0.00035	-0.00001	-0.00025	0.00084
Bootstrap p -value	0.857	0.995	0.905	0.635
<i>Panel C — full firm-level panel (WB index)</i>				
EP depth $\times$ Green: coefficient	-0.00226	-0.00090	-0.00185	-0.00196
Bootstrap p -value	0.686	0.833	0.673	0.611
EP depth $\times$ Dirty: coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p -value	0.185	0.176	0.049	0.035
<i>Panel D — full firm-level panel (TREND index)</i>				
EP count $\times$ Green: coefficient	-0.00010	-0.00011	-0.00012	-0.00039
Bootstrap p -value	0.931	0.919	0.920	0.703
EP count $\times$ Dirty: coefficient	-0.00089	-0.00101	-0.00140	-0.00140
Bootstrap p -value	0.177	0.142	0.069	0.082

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control; (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control. Wild cluster bootstrap ($B = 9,999$), clustered by destination. Panels C and D computed via `boottest` in Stata on residualized data (Frisch–Waugh).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 17: Permutation test: 1,000 random reassignments of environmental content

	(1)	(2)	(3)	(4)
<i>Panel A — WB index</i>				
EP depth $\times$ Green: observed coefficient	-0.00457	0.00173	-0.00433	-0.00675
Permutation p -value	0.597	0.897	0.480	0.465
Valid permutations	1,000	1,000	1,000	1,000
EP depth $\times$ Dirty: observed coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Permutation p -value	0.278	0.146	0.146	0.384
Valid permutations	1,000	1,000	1,000	1,000
<i>Panel B — TREND index</i>				
EP count $\times$ Green: observed coefficient	0.00181	0.00178	0.00161	0.00017
Permutation p -value	0.160	0.468	0.316	0.929
Valid permutations	1,000	1,000	1,000	1,000
EP count $\times$ Dirty: observed coefficient	0.00035	-0.00001	-0.00025	0.00084
Permutation p -value	0.817	0.996	0.904	0.788
Valid permutations	1,000	1,000	1,000	1,000

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control; (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control. EP profiles (depth and entry years) randomly reassigned among the treated destinations (23 in column 1, 25 in column 2, 22 in column 3 excluding Timor-Leste under DESTA, and 24 in column 4; $R = 1,000$); the p -value is the share of reassignments producing a coefficient at least as large in absolute value as the observed one. The 13 distinct EP profiles make the permutation distribution discrete.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

D Wild-cluster-bootstrap coefficient matrix

Table 18 reports the wild-cluster-bootstrap p -value for the dirty-goods coefficient across all four sample–depth combinations (two Hong Kong and Macao stances $\times$ two depth controls), on both the collapsed panel and the full firm-level panel. Using the same bootstrap procedure for all cells makes the eight p -values directly comparable, and the pattern they show is the one discussed in Section 5.3: the point estimate is uniformly negative, while significance tracks the choice of depth control rather than the sample.

Table 18: Summary: dirty-product coefficient across four specification variants, with common inference method

	(1)	(2)	(3)	(4)
Collapsed panel: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p -value	0.072	0.005	0.047	0.195
<i>Full panel</i> : coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p -value	0.185	0.176	0.049	0.035

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control; (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control. Coefficient on $EP_{dt}^{WB} \times b_p$. All p -values are wild cluster bootstrap ($B = 9,999$), clustered by destination.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

E Saturation ladder

Tables 19 and 20 report the full version of the ladder summarised in Table 6, for all three outcomes and for both the baseline specification without product-type interactions and the interacted specification. Two caveats apply to reading across columns. First, iterative singleton removal differs across fixed-effect structures, so the samples are not nested: N ranges from 22.9 million ($fpt + fpd$) to 35.6 million ($fpt + pd$), and the comparison therefore mixes the effect of the fixed effects with the effect of the sample. Second, none of the four structures in these tables includes firm–destination–year effects; the reference specification is reached only in column (5) of Table 6. The structural argument for why a destination–year intercept absorbs environmental depth entirely, and why the design is therefore built on composition rather than levels, is made in Section 4 and does not rest on these estimates.

Table 19: Saturation ladder — WB EP depth (EP_{dt}^{WB})

	Baseline				Interaction (g_p, b_p)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A — $\ln x_{fpdt}^{val}$</i>								
EP_{dt}^{WB}	0.0031 (0.0025)	0.0046* (0.0027)	0.0001 (0.0033)	0.0009 (0.0022)				
$EP_{dt}^{WB} \times g_p$					0.0018 (0.0050)	0.0012 (0.0027)	0.0060 (0.0050)	0.0050 (0.0037)
$EP_{dt}^{WB} \times b_p$					-0.0037 (0.0150)	-0.0015 (0.0027)	0.0058* (0.0031)	0.0064** (0.0027)
Observations	29,477,365	35,607,036	22,927,402	29,473,145	29,477,365	35,607,036	22,927,402	29,473,145
Adj. R^2	0.795	0.621	0.859	0.798	0.795	0.621	0.859	0.798
<i>Panel B — $\ln x_{fpdt}^{qua}$</i>								
EP_{dt}^{WB}	0.0042 (0.0028)	0.0033 (0.0028)	-0.0003 (0.0034)	0.0012 (0.0023)				
$EP_{dt}^{WB} \times g_p$					0.0023 (0.0037)	0.0032 (0.0026)	0.0061 (0.0060)	0.0053 (0.0038)
$EP_{dt}^{WB} \times b_p$					-0.0025 (0.0131)	-0.0006 (0.0028)	0.0048 (0.0031)	0.0058** (0.0025)
Observations	29,259,694	35,355,735	22,748,753	29,255,449	29,259,694	35,355,735	22,748,753	29,255,449
Adj. R^2	0.871	0.749	0.911	0.873	0.871	0.749	0.911	0.873
<i>Panel C — $\ln x_{fpdt}^{uv}$</i>								
EP_{dt}^{WB}	-0.0010* (0.0005)	0.0014*** (0.0003)	0.0005*** (0.0002)	-0.0002 (0.0003)				
$EP_{dt}^{WB} \times g_p$					0.0001 (0.0016)	-0.0019 (0.0017)	0.0002 (0.0016)	0.0003 (0.0008)
$EP_{dt}^{WB} \times b_p$					-0.0012 (0.0026)	-0.0014** (0.0006)	0.0005 (0.0006)	0.0008 (0.0006)
Observations	29,259,686	35,355,728	22,748,748	29,255,441	29,259,686	35,355,728	22,748,748	29,255,441
Adj. R^2	0.956	0.932	0.973	0.957	0.956	0.932	0.973	0.957
θ_{fpd}	✓		✓	✓	✓		✓	✓
θ_{pt}				✓				✓
θ_{fpt}		✓	✓			✓	✓	
θ_{pd}		✓				✓		
θ_t	✓				✓			
Clusters (d)	231	232	231	231	231	232	231	231

Notes: Each column represents a different fixed-effects structure. Columns (1)–(4) estimate the baseline EP_{dt}^{WB} coefficient without product-type interactions. Columns (5)–(8) decompose the EP effect by interacting with environmental (g_p) and pollution-intensive (b_p) product classifications. Hong Kong and Macao excluded. All specifications include the non-environmental depth control (TD_{dt}) and its interactions (not shown). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Chinese Customs Office; World Bank DTA Database.

Table 20: Saturation ladder — TREND EP count (EP_{dt}^{TR})

	Baseline				Interaction (g_p, b_p)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A</i> — $\ln x_{fpdt}^{val}$								
EP_{dt}^{TR}	0.0004 (0.0005)	0.0010** (0.0005)	0.0002 (0.0006)	0.0003 (0.0004)				
$EP_{dt}^{TR} \times g_p$					-0.0013 (0.0010)	0.0010 (0.0009)	0.0021** (0.0009)	0.0018** (0.0008)
$EP_{dt}^{TR} \times b_p$					-0.0049*** (0.0015)	0.000029 (0.0007)	0.0002 (0.0011)	0.0009 (0.0008)
Observations	29,477,365	35,607,036	22,927,402	29,473,145	29,477,365	35,607,036	22,927,402	29,473,145
Adj. R^2	0.795	0.621	0.859	0.798	0.795	0.621	0.859	0.798
<i>Panel B</i> — $\ln x_{fpdt}^{qua}$								
EP_{dt}^{TR}	0.0005 (0.0005)	0.0006 (0.0005)	0.000047 (0.0006)	0.0003 (0.0005)				
$EP_{dt}^{TR} \times g_p$					-0.0007 (0.0009)	0.0013 (0.0009)	0.0025** (0.0010)	0.0017** (0.0008)
$EP_{dt}^{TR} \times b_p$					-0.0034*** (0.0012)	0.0002 (0.0007)	0.0002 (0.0011)	0.0008 (0.0007)
Observations	29,259,694	35,355,735	22,748,753	29,255,449	29,259,694	35,355,735	22,748,753	29,255,449
Adj. R^2	0.871	0.749	0.911	0.873	0.871	0.749	0.911	0.873
<i>Panel C</i> — $\ln x_{fpdt}^{uv}$								
EP_{dt}^{TR}	-0.0001 (0.0001)	0.0004*** (0.0001)	0.0001*** (0.000047)	0.000022 (0.000048)				
$EP_{dt}^{TR} \times g_p$					-0.0004 (0.0003)	-0.0003 (0.0003)	-0.0002 (0.0003)	0.0002 (0.0002)
$EP_{dt}^{TR} \times b_p$					-0.0015*** (0.0003)	-0.0003* (0.0001)	-0.000005 (0.0001)	0.0002 (0.0001)
Observations	29,259,686	35,355,728	22,748,748	29,255,441	29,259,686	35,355,728	22,748,748	29,255,441
Adj. R^2	0.956	0.932	0.973	0.957	0.956	0.932	0.973	0.957
θ_{fpd}	✓		✓	✓	✓		✓	✓
θ_{pt}				✓				✓
θ_{fpt}		✓	✓			✓	✓	
θ_{pd}		✓				✓		
θ_t	✓				✓			
Clusters (d)	231	232	231	231	231	232	231	231

Notes: Each column represents a different fixed-effects structure. Columns (1)–(4) estimate the baseline EP_{dt}^{TR} coefficient without product-type interactions. Columns (5)–(8) decompose the EP effect by interacting with environmental (g_p) and pollution-intensive (b_p) product classifications. Hong Kong and Macao excluded. All specifications include the non-environmental depth control (TD_{dt}) and its interactions (not shown). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

F Dynamics: event study and destination-specific trends

Table 21 tabulates the two-way fixed-effects event study plotted in Figure 4. Pre-treatment coefficients are individually indistinguishable from zero on both margins. The negative coefficient on the green interaction at $t = 5$ should be read with the caution its position invites: the panel ends in 2015, so that bin covers the thinnest treated subsample and is identified by the earliest cohorts alone. The binary event studies do not reproduce the negative dirty interaction of the dose specification: the post-entry dirty coefficients are positive in Table 21 and in the Sun–Abraham aggregate (+0.073). The negative dose coefficient is therefore a contrast across agreements of different depth rather than a decline after entry within agreements; Appendix N returns to this point.

Table 21: Event study (TWFE) — EP leads and lags interacted with product type

Period	Baseline sample		DESTA-coverage sample	
	$\mathbf{1}[t] \times g_p$ (1)	$\mathbf{1}[t] \times b_p$ (2)	$\mathbf{1}[t] \times g_p$ (3)	$\mathbf{1}[t] \times b_p$ (4)
<i>Pre-treatment leads (parallel trends test)</i>				
$t \leq -6$	−0.044 (0.045)	0.027 (0.043)	−0.044 (0.045)	0.027 (0.043)
$t = -5$	−0.005 (0.038)	−0.017 (0.025)	−0.005 (0.038)	−0.017 (0.025)
$t = -4$	0.017 (0.027)	0.036 (0.027)	0.017 (0.027)	0.036 (0.027)
$t = -3$	−0.008 (0.020)	0.002 (0.028)	−0.008 (0.020)	0.002 (0.028)
$t = -2$	0.006 (0.019)	0.020 (0.039)	0.006 (0.019)	0.020 (0.039)
<i>Reference period</i>				
$t = -1$	0	0	0	0
<i>Post-treatment lags</i>				
$t = 0$	0.007 (0.014)	0.027 (0.027)	0.007 (0.014)	0.027 (0.027)
$t = 1$	0.010 (0.020)	0.017 (0.028)	0.011 (0.020)	0.017 (0.028)
$t = 2$	−0.020 (0.024)	0.021 (0.033)	−0.020 (0.024)	0.021 (0.033)
$t = 3$	−0.038 (0.026)	0.020 (0.034)	−0.038 (0.026)	0.020 (0.034)
$t = 4$	−0.039 (0.033)	0.021 (0.035)	−0.040 (0.033)	0.021 (0.035)
$t \geq 5$	−0.084** (0.034)	−0.032 (0.043)	−0.084** (0.034)	−0.032 (0.043)
Observations	3,681,023	3,681,023	3,677,333	3,677,333
Clusters (d)	228	228	228	228

Notes: TWFE event study on the collapsed panel. Binary treatment: $\mathbf{1}[\text{rel. time} = t]$ equals one for treated destinations t years from PTA entry into force, zero for never-treated. Coefficients: $\mathbf{1}[\text{rel. time} = t] \times g_p$ (cols. 1, 3) and $\mathbf{1}[\text{rel. time} = t] \times b_p$ (cols. 2, 4), estimated jointly; no depth control. $t = -1$ is the omitted reference period; endpoint bins accumulate (≤ -6 , $\geq +5$). FE: $pd + dt + pt$; weights: cell counts. Columns (3)–(4) drop Timor-Leste (not covered by DESTA). Standard errors clustered at the destination level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database.

G Sun–Abraham event study

Figure 5 and Table 22 report the interaction-weighted (L. Sun and Abraham, 2021) event study of the green and dirty versus neutral differential, estimated on the destination-year composition gap.

A note on standard errors. The cohort shares used to aggregate group-specific effects are themselves estimated from the data; L. Sun and Abraham (2021) prescribe including that estimation uncertainty in the variance. The STATA implementation does so; the R implementation treats the shares as fixed weights.¹⁹ The two agree on coefficients to 15 significant figures but disagree on standard errors. The degree of disagreement is informative: where a single cohort identifies a period (so there are no shares to estimate), the ratio of the two standard errors is exactly 1.00; where many cohorts contribute and their effects are heterogeneous, the ratio reaches 3 to 4. The standard errors reported here are the STATA ones, which reflect the full estimation uncertainty.

With these standard errors, no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on the green or dirty margin. The ATT is -0.042 ($p = 0.27$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap. Any apparent pre-trend in the R output (the lead at $t = -6$ had $p = 0.001$ with fixed-weight standard errors) reflects the treatment of cohort-share uncertainty, not a genuine pre-trend anomaly.

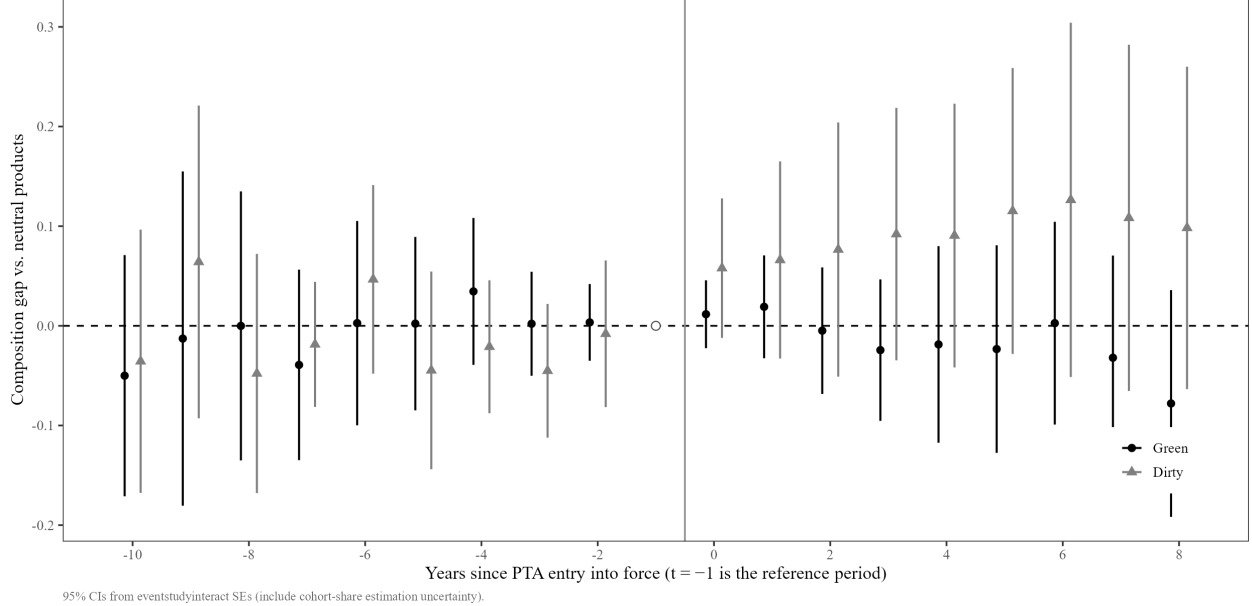
¹⁹The STATA estimates use `eventstudyinteract` (); the R estimates use the Sun–Abraham estimator in `fixest::sunab` (Bergé, 2018).

Table 22: Sun–Abraham interaction-weighted event study: composition gap estimates

Years since entry	Green – neutral gap		Dirty – neutral gap	
	Coeff.	<i>p</i>	Coeff.	<i>p</i>
–10	–0.0500	0.419	–0.0356	0.598
–9	–0.0128	0.881	0.0641	0.424
–8	–0.0001	0.999	–0.0478	0.436
–7	–0.0392	0.422	–0.0186	0.561
–6	0.0027	0.959	0.0466	0.335
–5	0.0022	0.961	–0.0447	0.378
–4	0.0345	0.359	–0.0210	0.538
–3	0.0021	0.938	–0.0451	0.189
–2	0.0034	0.861	–0.0080	0.832
+0	0.0116	0.504	0.0579	0.107
+1	0.0191	0.470	0.0661	0.192
+2	–0.0049	0.881	0.0765	0.241
+3	–0.0244	0.501	0.0921	0.156
+4	–0.0187	0.711	0.0906	0.181
+5	–0.0233	0.661	0.1153	0.117
+6	0.0027	0.959	0.1264	0.165
+7	–0.0320	0.541	0.1083	0.223
+8	–0.0779	0.181	0.0983	0.235
Aggregated ATT	–0.0421	0.269	0.0729	0.276

Notes: Estimated via `eventstudyinteract` in Stata (L. Sun and Abraham, 2021). Dependent variable: destination–year composition gap (mean log exports of green or dirty products minus mean log exports of neutral products, within destination–year cells). Standard errors propagate estimation uncertainty in cohort shares, as prescribed by L. Sun and Abraham (2021); `fixest::sunab` in R treats the shares as known weights and yields narrower standard errors — the ratio reaches 3–4 in periods identified by many discordant cohorts. Coefficients agree to 15 significant figures across implementations. Periods outside the $[-10, +8]$ window are estimated (t down to -15 and up to $+13$) but not displayed; the endpoints shown are individual period coefficients, not accumulated bins. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 5: Sun–Abraham interaction-weighted event study: green and dirty vs. neutral gap at the destination–year level. 95% confidence bands account for estimation uncertainty in cohort shares. The window $[-10, +8]$ is shown; the estimates outside it rest on one or two cohorts each.



H Destination-specific trends and pre-trend tests

Table 23 reports the two trend specifications discussed in Section 5.6. Panels A and A' add a linear trend for each destination interacted with product type, estimated over the full sample period; Panels B and B' estimate destination-specific slopes on pre-treatment years only and project them forward before re-estimating on the detrended outcome.

The only green coefficient in this appendix that survives the wild cluster bootstrap appears here: the TREND green interaction under full-period trends, at -0.0022 with bootstrap $p = 0.015$. Two considerations argue against taking it at face value. Trends estimated over the full sample period absorb both pre- and post-treatment variation and can reverse the sign of treatment coefficients when cohort dynamics are present (Wolfers, 2006), and the late negative drift these trends absorb is precisely the early-cohort artifact that the Sun–Abraham estimator identifies as noise. Panels A and B re-estimate the interactions after removing destination-specific trends fitted on pre-entry years only. The TREND green interaction becomes $+0.0073$ (bootstrap $p = 0.19$), the WB dirty interaction $+0.063$ ($p = 0.61$): the detrended estimates are imprecise and, for dirty, of the opposite sign, because most cohorts offer only two to four pre-entry years on which to fit a slope. The variant therefore cannot adjudicate; the Wolfers argument above is what carries the reading of the full-period trend result.

Table 23: Destination-specific trends and pre-trend tests (collapsed panel)

	(1)	(2)	(3)	(4)
<i>Panel A — destination×product-type linear trends (WB index)</i>				
EP depth × Green	-0.00510 (0.00360) [0.494]	-0.00200 (0.00436) [0.808]	0.00084 (0.00434) [0.896]	-0.00006 (0.00441) [0.995]
EP depth × Dirty	-0.00825* (0.00423) [0.275]	-0.00927** (0.00414) [0.249]	-0.01030*** (0.00239) [0.086]	-0.00930*** (0.00204) [0.037]
<i>Panel A' — same specification, TREND index</i>				
EP count × Green	-0.00217*** (0.00062) [0.015]	-0.00214*** (0.00061) [0.018]	-0.00111** (0.00053) [0.218]	-0.00153*** (0.00049) [0.074]
EP count × Dirty	-0.00137 (0.00115) [0.355]	-0.00142 (0.00114) [0.339]	-0.00231*** (0.00066) [0.048]	-0.00164** (0.00064) [0.109]
<i>Panel B — pre-trend-detrended outcome (WB index)</i>				
EP × Green, detrended outcome	0.01732 (0.01295)	0.00402 (0.01678)	0.01719* (0.00997)	0.02218 (0.01346)
Bootstrap <i>p</i> -value	0.678	0.705	0.543	0.497
EP × Dirty, detrended outcome	0.06292 (0.04393)	0.06237 (0.03983)	0.07109** (0.03161)	0.07056** (0.03159)
Bootstrap <i>p</i> -value	0.610	0.515	0.416	0.410
<i>Panel B' — pre-trend-detrended outcome (TREND index)</i>				
EP × Green, detrended outcome	0.00728* (0.00438)	0.00707 (0.00441)	0.00775* (0.00428)	0.00944** (0.00389)
Bootstrap <i>p</i> -value	0.189	0.199	0.173	0.090
EP × Dirty, detrended outcome	0.01512** (0.00728)	0.01512** (0.00721)	0.01839*** (0.00596)	0.01703*** (0.00613)
Bootstrap <i>p</i> -value	0.293	0.301	0.164	0.243

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control; (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control. Panels A/A': linear destination×product-type trends estimated over the full sample period. Panels B/B': for each destination, a linear trend of the green and dirty gaps is estimated on pre-entry years only (all years for never-treated destinations), projected over the full period and subtracted from the outcome; the rows report the EP interactions re-estimated on the detrended outcome. The pre-entry slopes themselves are not reported. Standard errors in parentheses; wild cluster bootstrap *p*-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

I Sub-index estimates on the collapsed panel

Table 24 reports the sub-index estimates on the collapsed panel, the counterpart to Table 13 in the main text. Each column is a separate regression replacing the aggregate index with one sub-index; the coefficients are not partial effects. The contrast between the two tables follows the contrast between the two panels on the aggregate index: coefficients that reach significance on the collapsed panel — Green Liberalization and Green Market Access interacted with dirty products — are close to zero on the full panel, which locates them in the between-firm channel that θ_{fdt} absorbs. See Section 5.5 for interpretation.

Table 24: Collapsed panel — sub-index decomposition

	WB sub-indices		TREND sub-indices				
	(1) Green Lib.	(2) Enf. DSM	(3) Green Mkt.	(4) Enf. DSM	(5) Hard	(6) Soft	(7) Reg. Space
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>							
$EP_{\text{sub}} \times g_p$	-0.0115 (0.0879)	-0.0048 (0.0394)	-0.0271 (0.0303)	0.0042 (0.0148)	0.0022 (0.0047)	0.0132 (0.0121)	0.0242** (0.0099)
$EP_{\text{sub}} \times b_p$	-0.0876* (0.0483)	0.0067 (0.0525)	-0.0491** (0.0243)	-0.0052 (0.0140)	-0.0013 (0.0037)	0.0019 (0.0108)	0.0225*** (0.0086)
θ_{pd}	✓	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023
Clusters (d)	228	228	228	228	228	228	228
<i>Panel B — DESTA depth control</i>							
$EP_{\text{sub}} \times g_p$	-0.0141 (0.0892)	-0.0105 (0.0280)	-0.0290 (0.0288)	0.0032 (0.0160)	0.0017 (0.0047)	0.0132 (0.0131)	0.0149 (0.0104)
$EP_{\text{sub}} \times b_p$	-0.1001** (0.0398)	-0.0197 (0.0343)	-0.0543*** (0.0187)	-0.0086 (0.0115)	-0.0023 (0.0029)	-0.0015 (0.0101)	0.0092 (0.0075)
θ_{pd}	✓	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333
Clusters (d)	228	228	228	228	228	228	228

Notes: Collapsed panel (HS6–destination–year cells), excluding Hong Kong and Macao. Dependent variable: mean log export value ($\bar{y}_{pdt}$), weighted by cell size. Each column estimates a separate model replacing the aggregate EP index with one sub-index; the non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. Fixed effects: product-destination (θ_{pd}), destination-year (θ_{dt}), product-year (θ_{pt}). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

WB sub-indices: (1) Green Liberalization; (2) Enforcement & DSM. TREND sub-indices: (3) Green Market Access; (4) Enforcement & DSM; (5) Hard obligations; (6) Soft obligations; (7) Regulatory Space.

Table 25: Composition of EP sub-indices

Sub-index	Source	Content
<i>Trade-mechanism sub-indices</i>		
Green Liberalization	WB	Liberalization of trade in environmental goods and services (WB_10: tariff reductions and removal of non-tariff barriers on green products)
Standards Regression	Non-WB	Commitments not to lower domestic environmental standards to attract trade or investment (WB_2, WB_8, WB_9); collinear with Green Liberalization in-sample, not estimated separately
Green Market Access	TREND	Provisions facilitating market access for environmental goods (X7_01_01, X7_01_02_01–02, X8_09_04)
<i>Enforcement sub-indices</i>		
Enforcement & DSM (WB)	WB	Dispute resolution and enforcement mechanisms specific to the environmental chapter (WB_13–WB_16)
Enforcement & DSM (TREND)	TREND	Environmental dispute settlement and enforcement procedures (X5_*, X11_*, X12_*, X13_*)
<i>Cooperation and regulatory sub-indices (no direct trade mechanism)</i>		
Hard obligations	TREND	Obligations stated in binding terms (X2_*, X5_*, X10_*, X14_*, net of the soft-obligation provisions)
Soft obligations	TREND	Cooperation and best-endeavour language (X1_*, X7_09, X5_01_02)
Regulatory Space	TREND	Regulatory autonomy and policy-space clauses (X1_07–X1_09, X8_*)

Notes: All sub-indices are constructed here from the raw provision-level codings in the WB DTA and TREND databases; they are not variables native to either database. “Trade-mechanism” sub-indices contain provisions with an identifiable channel to trade flows. “Cooperation and regulatory” sub-indices cover provisions that, while environmentally relevant, lack a direct mechanism through which they could shift the product composition of exports. Each sub-index is a count of the relevant provisions present in a given agreement. Source: World Bank DTA Database; TREND Database.

J Hong Kong and Macao

Hong Kong and Macao are excluded from the estimation samples in the main text. Both are covered by Closer Economic Partnership Arrangements with the mainland, and both function as entrepôts: a large share of what China ships to them is re-exported onward (Feenstra and Hanson, 2004), so a shipment recorded as an export to Hong Kong is frequently not a shipment to a Hong Kong buyer. Including them adds two treated destinations whose recorded flows are a composite of final demand and transit.

Tables 26 and 27 re-estimate both main specifications with the two territories included. On the green margin nothing changes. On the dirty margin the collapsed-panel coefficient roughly doubles under the World Bank depth control, to -0.0189 with a bootstrap p -value of 0.005, while the permutation p -value stays at 0.146 — the same divergence between the two tests documented in Section 5.3, in a sharper form. The leave-one-out rows for the two territories in Table 9 show that dropping Hong Kong alone moves the coefficient from -0.0189 to -0.0129 : of the 0.0070 increase relative to the baseline of -0.0119 , Hong Kong accounts for 0.0060, and Macao for almost none (-0.0183 when dropped). This is the reason the entrepôt economies are held out of the baseline rather than a robustness footnote.

Table 26: Full panel including Hong Kong and Macao — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0036)	−0.0001 (0.0038)	−0.0004 (0.0013)	−0.0001 (0.0010)	−0.0001 (0.0011)	−0.0001 (0.0003)
Asymptotic p	[0.804]	[0.984]	[0.734]	[0.913]	[0.960]	[0.847]
Bootstrap p	[0.833]	[0.992]	—	[0.919]	[0.965]	—
$EP_{dt} \times b_p$	−0.0041 (0.0024)	−0.0036 (0.0031)	−0.0008 (0.0010)	−0.0010 (0.0006)	−0.0008 (0.0008)	−0.0001 (0.0004)
Asymptotic p	[0.094]	[0.236]	[0.451]	[0.112]	[0.321]	[0.836]
Bootstrap p	[0.176]	[0.422]	—	[0.142]	[0.340]	—
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	23,560,110	23,334,037	23,334,027	23,560,110	23,334,037	23,334,027
Clusters (d)	227	227	227	227	227	227
R^2	0.871	0.910	0.963	0.871	0.910	0.963
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0020 (0.0035)	−0.0016 (0.0032)	−0.0004 (0.0019)	−0.0004 (0.0009)	−0.0003 (0.0009)	−0.0001 (0.0004)
Asymptotic p	[0.572]	[0.617]	[0.824]	[0.675]	[0.716]	[0.723]
Bootstrap p	[0.611]	[0.713]	—	[0.703]	[0.754]	—
$EP_{dt} \times b_p$	−0.0057** (0.0027)	−0.0076** (0.0031)	0.0008 (0.0010)	−0.0014* (0.0007)	−0.0017* (0.0009)	0.0002 (0.0003)
Asymptotic p	[0.034]	[0.015]	[0.419]	[0.052]	[0.058]	[0.524]
Bootstrap p	[0.035]	[0.012]	—	[0.082]	[0.083]	—
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	23,381,204	23,155,803	23,155,793	23,381,204	23,155,803	23,155,793
Clusters (d)	227	227	227	227	227	227
R^2	0.871	0.911	0.963	0.871	0.911	0.963

Notes: Full firm-level panel, including Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv} = \ln x_{fpdt}^{val} - \ln x_{fpdt}^{qua}$). Quantity and unit-value regressions are estimated on the subset of observations with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets; “—” indicates bootstrap not available for this outcome.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 27: Collapsed panel including Hong Kong and Macao — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})	TREND EP count (EP_{dt}^{TR})
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{val}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>		
$EP_{dt} \times g_p$	0.0017 (0.0091)	0.0018 (0.0018)
Asymptotic p	[0.849]	[0.337]
Bootstrap p	[0.825]	[0.414]
Permutation p	[0.897]	[0.468]
$EP_{dt} \times b_p$	-0.0189*** (0.0061)	-0.00001 (0.0015)
Asymptotic p	[0.002]	[0.995]
Bootstrap p	[0.005]	[0.995]
Permutation p	[0.146]	[0.996]
θ_{pd}	✓	✓
θ_{dt}	✓	✓
θ_{pt}	✓	✓
Observations	3,769,187	3,769,187
Clusters (d)	230	230
<i>Panel B — DESTA depth control</i>		
$EP_{dt} \times g_p$	-0.0067 (0.0048)	0.0002 (0.0020)
Asymptotic p	[0.157]	[0.931]
Bootstrap p	[0.484]	[0.949]
Permutation p	[0.465]	[0.929]
$EP_{dt} \times b_p$	-0.0082 (0.0043)	0.0008 (0.0015)
Asymptotic p	[0.055]	[0.575]
Bootstrap p	[0.195]	[0.635]
Permutation p	[0.384]	[0.788]
θ_{pd}	✓	✓
θ_{dt}	✓	✓
θ_{pt}	✓	✓
Observations	3,759,733	3,759,733
Clusters (d)	230	230

Notes: Collapsed panel (product×destination×period cells, analytic weights), including Hong Kong and Macao. Dependent variable: log export value ($\ln x_{fpdt}^{val}$). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. Permutation p -values (treated-only design, $R = 1,000$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

K Full robustness battery with standard errors

Table 28 reproduces the full-panel robustness specifications with standard errors and asymptotic p -values for both the WB and TREND indices. The variants are: adding product–destination–year controls for tariffs, market concentration and antidumping exposure; dropping the eleven destinations covered by the ASEAN cohort (the ten members of the China–ASEAN agreement plus Timor-Leste, which the customs data group with them); re-including Hong Kong and Macao; restricting to products with common support across agreement and non-agreement destinations; and replacing the dependent variable with the green share of the firm’s export basket to that destination.

A note on the tariff control. The tariff variable is the applied duty recorded in the customs data. It varies across partner destinations for the same HS6–year in 97.9 percent of product–year cells, so it is a destination-specific tariff faced by Chinese exporters rather than an import duty levied by China. Its average does not fall after PTA entry for partner destinations, which indicates that it records the MFN rate rather than the preferential rate; the preferential rate was not obtainable from these data. Two features limit what this costs. The tariff enters only this table and the CEM matching, not the headline estimates; and any tariff change common to all products shipped to a destination in a year is absorbed by θ_{fdt} , so only product-specific variation contributes. If PTA negotiations cut tariffs more on green goods and the omitted preferential rate captured that cut, the effect would load onto the green interaction with a positive sign. The estimate is zero, so the omission works against finding a positive effect rather than toward it.

The dirty margin is exposed in the opposite direction. Pollution-intensive products — steel, chemicals, refined petroleum, paper — are the sectors most often placed on partners’ sensitive lists, where tariffs fall later and less: under the ASEAN–China agreement up to 400 tariff lines per member remained above 20 percent until 2012 and at 50 percent until 2015. If partners cut tariffs less on dirty than on neutral products, dirty exports grow less than neutral ones after entry into force and the dirty interaction turns negative with no role for the environmental chapter. The non-environmental depth control cannot separate this, because within the ASEAN bloc environmental and non-environmental depth are constant. We cannot rule this channel out with the data at hand; it is the most economical alternative reading of the collapsed-panel dirty coefficient, and it is consistent with the fact that the coefficient is carried by the medium-depth ASEAN cohort (Appendix N).

Table 28: Robustness on the full panel: sample, control, and dependent-variable variants

Variant	× Green	× Dirty	Observations	R ²
<i>Panel A — WB index</i>				
Additional controls ^a	-0.00024 (0.00296) [0.935]	-0.00600** (0.00259) [0.022]	19,286,638	0.8736
Excluding the ASEAN cohort (11 destinations)	-0.00253 (0.00360) [0.482]	-0.00438* (0.00223) [0.051]	19,236,090	0.8719
Including Hong Kong and Macao	-0.00090 (0.00360) [0.804]	-0.00409* (0.00243) [0.094]	23,560,110	0.8706
Common-support products only	-0.00226 (0.00392) [0.566]	-0.00435* (0.00223) [0.052]	21,519,197	0.8734
PTA partners only, deep vs. shallow	-0.00222 (0.00351) [0.534]	-0.00344 (0.00289) [0.246]	5,262,293	0.8839
Green share in firm's export basket	-0.00010 (0.00015) [0.502]		13,345,132	0.8510
<i>Panel B — TREND index</i>				
Additional controls ^a	+0.00039 (0.00091) [0.669]	-0.00085 (0.00068) [0.209]	19,286,638	0.8736
Excluding the ASEAN cohort (11 destinations)	-0.00102 (0.00102) [0.318]	-0.00183** (0.00072) [0.012]	19,236,090	0.8719
Including Hong Kong and Macao	-0.00011 (0.00100) [0.913]	-0.00101 (0.00063) [0.112]	23,560,110	0.8706
Common-support products only	-0.00010 (0.00099) [0.924]	-0.00090 (0.00062) [0.153]	21,519,197	0.8734
PTA partners only, deep vs. shallow	-0.00036 (0.00100) [0.721]	-0.00061 (0.00101) [0.549]	5,262,293	0.8839
Green share in firm's export basket	-0.00006** (0.00003) [0.044]		13,345,132	0.8510

Notes: Full firm-level panel, excl. HK/Macao, TotalDepth control. Fixed effects and clustering as in the main specification unless noted. ^aAdditional controls: applied tariff, product-level exporter concentration, antidumping exposure. ASEAN cohort: the ten China–ASEAN members plus Timor-Leste. Common support: only products exported to both agreement and non-agreement destinations. Green share in basket: dependent variable is the share of green-product revenue to that destination; FE: firm×destination and year. Standard errors in parentheses, *p*-values in brackets. * *p* < 0.10, ** *p* < 0.05, *** *p* < 0.01.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

L Alternative depth controls

Environmental depth and general agreement depth are strongly correlated, and Section 5.2 shows that the dirty coefficient’s precision depends on which control is used. Table 29 varies the control across four alternatives on the collapsed panel: none, aggregate World Bank TotalDepth, a version restricted to the 14 non-environmental areas most correlated with EP content, and the DESTA index. On the green margin the point estimate moves within a band of 0.0024 log points — narrower than a single standard error — stays negative throughout, and every interval contains zero. Dropping the depth control altogether moves the coefficient away from zero rather than toward it, which is what one expects if EP depth then also captures the effect of overall agreement depth. The green null does not rest on the choice of depth control.

Table 29: EP×green under alternative depth controls (collapsed panel, WB index)

Depth control	EP×green	Asymptotic 95% CI	<i>p</i>
None	−0.0057	[−0.0118, +0.0003]	0.07
TotalDepth (17 areas) <i>[baseline]</i>	−0.0046	[−0.0182, +0.0091]	0.51
TotalDepth targeted (14 areas)	−0.0033	[−0.0188, +0.0123]	0.68
DESTA index	−0.0043	[−0.0129, +0.0042]	0.32
Product–dest FE		Yes	
Dest–year FE		Yes	
Product–year FE		Yes	
Cells		3,681,023	
Clusters		228 destinations	

Notes: Collapsed panel, weighted by cell size. Asymptotic intervals. The targeted control drops three WB areas with zero variance in Chinese agreements (labour-market regulation, visa and asylum, subsidies). DESTA covers seven policy areas coded independently of the World Bank and excludes environmental provisions by construction.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database; DESTA.

M Entry-into-force timing

Section 4 codes a destination as treated from the calendar year of entry into force. Seven agreements enter in the second half of the year, and two of them — Australia and Korea, both 20 December 2015 — fall in the panel’s final year, so their treated year records eleven days of exposure. Table 30 re-estimates the collapsed-panel specification under six variants: the baseline; coding Korea and Australia’s 2015 treatment variables at their pre-entry (2014)

values (A1); doing the same for all seven second-half entrants (A2); dropping 2015 entirely (B); dropping Korea and Australia outright (C); and treating single-provision (depth-1) destinations as untreated, since a lone 1975 Bangkok Agreement provision is a weak claim to a substantive environmental chapter (D). The WB dirty coefficient is estimated imprecisely under every variant that touches the two 2015 entrants (A1, A2, B, C): the point estimate loses its sign and its standard error roughly triples relative to the baseline, consistent with Section 4’s observation that Korea and Australia carry a large share of the identifying variation on that margin. The TREND dirty coefficient is small and imprecise in the baseline and in D, but turns marginally significant ($p \approx 0.06$ – 0.08) and positive in A1, A2, B and C — the opposite sign from the WB result on the same subsample, which is a caution against reading either index’s dirty coefficient as robust to how the two 2015 entrants are coded. The WB green coefficient is never distinguishable from zero. The TREND green coefficient tells a different story: an imprecise zero in the baseline and D ($p = 0.32$ and 0.24), it becomes positive and precisely estimated ($p < 0.001$) in every variant that recodes or removes the two 2015 entrants (A1, A2, B, C). This is the reverse of what an omitted pre-existing trend would produce, and it says that Korea and Australia’s very deep, very late chapters pull the TREND green estimate toward zero in the baseline rather than driving it; the coefficient the timing variants recover is closer to a per-provision effect of the earlier, shallower chapters.

Table 30: Entry-into-force timing: robustness of the collapsed-panel estimates

Variant	WB index		TREND index	
	EP×green	EP×dirty	EP×green	EP×dirty
Baseline	−0.0046 (0.0070)	−0.0119*** (0.0030)	0.0018 (0.0018)	0.0004 (0.0016)
A1: Korea & Australia 2015 coded pre-EIF	0.0175 (0.0124)	−0.0059 (0.0094)	0.0052*** (0.0012)	0.0030* (0.0016)
A2: all second-half-entry years coded pre-EIF	0.0085 (0.0120)	−0.0088 (0.0117)	0.0050*** (0.0012)	0.0032* (0.0017)
B: drop 2015	0.0100 (0.0145)	−0.0064 (0.0113)	0.0045*** (0.0012)	0.0029* (0.0016)
C: drop Korea and Australia entirely	0.0174 (0.0123)	−0.0055 (0.0094)	0.0052*** (0.0013)	0.0030* (0.0016)
D: single-provision (depth 1) destinations coded untreated	−0.0041 (0.0070)	−0.0120*** (0.0031)	0.0021 (0.0018)	0.0005 (0.0016)

Notes: Collapsed panel, WB total depth control, destination clustering. A1 and A2 recode the treatment variables of the affected destination–years to their pre-entry (previous-year) values rather than dropping observations. Asymptotic standard errors in parentheses. Cluster counts and observation counts are within a few hundred cells of the baseline (3,681,023 cells, 228 clusters) except B (3,344,592 cells, 227 clusters, 2015 dropped) and C (3,573,947 cells, 226 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

N Dose bins

Section 4 treats environmental depth as a continuous dose, which imposes linearity on the response rather than testing it. Table 31 replaces the continuous depth control with three indicators for the current-year dose — low (1–5 provisions), medium (6–7), and high (8 or more) — estimated against the never-treated destination-years. The bins are grumous: eleven of the 23 treated destinations sit at exactly six provisions (the ASEAN bloc plus Iceland), and only three destination-years exceed seven (Peru at 12, Switzerland at 14, Korea at 17 in 2015 only), so the high bin rests on one country per dose level and its standard error should be read as uninformative rather than as evidence of a large effect. On the green margin every bin coefficient is imprecise, and the one that clears the 10 percent threshold under the asymptotic standard error (the low bin, $+0.064$) does not survive the wild cluster bootstrap ($p = 0.165$), the same wedge the paper’s main results show throughout with few treated clusters. On the dirty margin the bins move in different directions — positive in the low bin, negative and larger in magnitude than the linear fit implies in the medium and high bins — so the negative linear dirty coefficient of Table 8 is better read as a contrast between the medium-depth ASEAN cohort and the rest of the sample than as a per-provision slope; neither bin individually reaches significance under the bootstrap. The dose-response evidence therefore does not support a stronger reading of either margin than the linear specification already gives.

Table 31: Dose bins: the current-dose response is not linear

Fascia (dose corrente)	Green			Dirty		
	Coef.	Implied by linear fit	p (WCB)	Coef.	Implied by linear fit	p (WCB)
Low (1–5, median 1)	0.0643* (0.0308)	−0.0046	0.165	0.0364 (0.0351)	−0.0119	0.666
Medium (6–7, median 6)	0.0403 (0.0721)	−0.0274	0.608	−0.1060 (0.0882)	−0.0712	0.284
High (8+, median 12)	0.2717 (0.1772)	−0.0548	—	−0.1700 (0.2325)	−0.1425	—
Countries	9			13		3 (high)
Country–years	83	131		9		

Notes: Collapsed panel, weighted by cell size, FE $pd + dt + pt$, clustered by destination (228 clusters, 3,681,023 cells for every row). Bins are defined on the current-year WB total depth (never-treated destination–years are the omitted category). “Implied by linear fit” is the baseline linear coefficient (Table 8, column WB) times the bin’s median depth; the gap between it and the bin coefficient is the departure from linearity the design is built to detect. Standard errors in parentheses are asymptotic. p (WCB): wild cluster bootstrap p -value (Rademacher weights, 9,999 replications), computed only for the low and medium bins, which have countries at more than one dose level within the bin; the high bin has one country per dose level (Peru 12, Switzerland 14, Korea 17, only in 2015) and a bootstrap adds no information the asymptotic standard error does not already convey. The low-bin green coefficient is significant under the asymptotic standard error ($p = 0.038$) but not under the wild cluster bootstrap ($p = 0.165$), the same wedge the main results show throughout with few treated clusters. * $p < 0.10$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

O APEC versus OECD product classification

Table 32 compares the estimates when the green product list is restricted to the 54 APEC Environmental Goods List codes (Sauvage, 2014) against the baseline using the 248 OECD CLEG codes. The APEC list is a strict subset of the OECD list, agreed politically in 2012 and much narrower in coverage. If the green null depended on which products are labelled environmental, the two columns would diverge; they do not.

Table 32: Alternative environmental-good definition: OECD list vs. APEC list

	OECD list (used in this paper)	APEC list (54 codes)
EP depth (WB) $\times$ Green	-0.00457 (0.00696) [0.512]	0.00501 (0.01266) [0.693]
EP count (TREND) $\times$ Green	0.00181 (0.00182) [0.320]	0.00319 (0.00208) [0.126]

Notes: Collapsed panel, excl. HK/Macao, TotalDepth control. The APEC Environmental Goods List (54 codes) replaces the OECD CLEG list (248 codes) used in the baseline. Standard errors in parentheses, p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

P CO₂ intensity as a continuous dirty measure

The binary dirty indicator groups HS6 products into five pollution-intensive sectors, and a continuous measure of emission intensity might tell a different story. We construct one from the replication data of Shapiro (2021), which reports China-specific CO₂ emission intensities for 47 EXIOBASE industries.²⁰ HS6 products are mapped to ISIC3 divisions through the same concordance used for the binary classification; unmatched codes, 9.5 percent of the panel, are assigned the sample mean. The resulting measure validates against the existing trichotomy without circularity: mean intensity is 0.0012 for dirty products, 0.0008 for neutral and 0.0006 for green, monotonic in the expected direction.

Table 33 re-estimates the collapsed-panel specification with the standardized intensity in place of the binary indicator. The interaction with the World Bank index is -0.0026 with a bootstrap p -value of 0.71; with the TREND index it is -0.0013 , asymptotically significant at $p = 0.011$ but with a bootstrap p -value of 0.063. The continuous measure fades under robust inference the same way the binary one does. It is tested on the collapsed panel only: re-estimating the interaction structure on the full panel would require substantial additional computation, and the collapsed-panel test is sufficient to establish that the continuous measure behaves like the binary indicator.

²⁰Shapiro (2021) documents that most countries' tariff structures implicitly subsidize pollution-intensive sectors and tax clean ones — a systematic environmental bias that environmental chapters are meant to work against.

Table 33: CO₂ intensity (Shapiro, 2021) — collapsed panel, triple-difference

	WB depth control (TD_{dt})		DESTA depth control	
	(1)	(2)	(3)	(4)
	Excl. HK/Macao	Incl. HK/Macao	Excl. HK/Macao	Incl. HK/Macao
<i>Panel A — WB EP depth (EP_{dt}^{WB})</i>				
$EP_{dt} \times g_p$	−0.0050 (0.0070)	0.0004 (0.0080)	−0.0025 (0.0049)	−0.0041 (0.0055)
Asymptotic p	[0.472]	[0.956]	[0.613]	[0.455]
Bootstrap p	[0.544]	[0.961]	[0.797]	[0.797]
$EP_{dt} \times \text{CO}_2$	−0.0026 (0.0057)	−0.0049 (0.0064)	0.0013 (0.0031)	0.0028 (0.0031)
Asymptotic p	[0.653]	[0.448]	[0.679]	[0.373]
Bootstrap p	[0.707]	[0.529]	[0.773]	[0.384]
<i>Panel B — TREND EP count (EP_{dt}^{TR})</i>				
$EP_{dt} \times g_p$	0.0009 (0.0017)	0.0009 (0.0017)	0.0011 (0.0018)	−0.00002 (0.0017)
Asymptotic p	[0.605]	[0.617]	[0.546]	[0.990]
Bootstrap p	[0.677]	[0.698]	[0.652]	[0.993]
$EP_{dt} \times \text{CO}_2$	−0.0013** (0.0005)	−0.0014** (0.0005)	−0.0008** (0.0004)	−0.0002 (0.0006)
Asymptotic p	[0.011]	[0.012]	[0.032]	[0.779]
Bootstrap p	[0.063]	[0.056]	[0.083]	[0.817]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	3,681,023	3,769,187	3,677,333	3,759,733
Clusters (d)	228	230	228	230

Notes: Collapsed panel (product×destination×year cells, analytic weights). The binary dirty indicator is replaced by a continuous product-level CO₂ emission intensity from Shapiro (2021), mapped to HS6 codes. $EP_{dt} \times g_p$ captures the differential effect on environmental goods; $EP_{dt} \times \text{CO}_2$ captures the marginal effect of a one-unit increase in CO₂ intensity. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times \text{CO}_2$ controls (not shown). Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database; Shapiro (2021).

Q Extended tables for the comparison-group subsamples

Tables 34, 35 and 36 report the three subsamples of Table 11 in full: three outcomes, two provision codings and two depth controls each. The synoptic table in the main text reports the export-value column of each under the wild cluster bootstrap; these give the quantity and unit-value columns and the asymptotic p -values alongside.

Q.1 C-prod-HS4

Q.2 CEM-matched destinations

Q.3 Deep versus shallow agreements

Table 34: C-prod-HS4 subsample — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0051)	−0.0019 (0.0043)	0.0005 (0.0007)	0.0005 (0.0011)	0.0001 (0.0010)	0.0002 (0.0004)
Asymptotic p	[0.858]	[0.665]	[0.471]	[0.680]	[0.906]	[0.689]
Bootstrap p	[0.860]	[0.662]	[0.421]	[0.730]	[0.916]	[0.713]
$EP_{dt} \times b_p$	0.0066 (0.0090)	0.0068 (0.0111)	−0.0017 (0.0070)	0.0024 (0.0066)	0.0033 (0.0059)	−0.0004 (0.0024)
Asymptotic p	[0.469]	[0.541]	[0.807]	[0.714]	[0.581]	[0.860]
Bootstrap p	[0.435]	[0.562]	[0.857]	[0.734]	[0.591]	[0.866]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,772,855	3,745,788	3,745,788	3,772,855	3,745,788	3,745,788
Clusters (d)	215	215	215	215	215	215
Adj. R^2	0.737	0.835	0.923	0.737	0.835	0.923
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0051)	−0.0032 (0.0042)	0.0012 (0.0012)	0.0004 (0.0013)	−0.0003 (0.0011)	0.0003 (0.0004)
Asymptotic p	[0.867]	[0.449]	[0.338]	[0.779]	[0.790]	[0.484]
Bootstrap p	[0.855]	[0.461]	[0.368]	[0.817]	[0.806]	[0.546]
$EP_{dt} \times b_p$	0.0033 (0.0168)	−0.0059 (0.0182)	0.0076 (0.0098)	0.0015 (0.0040)	−0.0006 (0.0032)	0.0022 (0.0025)
Asymptotic p	[0.846]	[0.747]	[0.440]	[0.719]	[0.854]	[0.383]
Bootstrap p	[0.850]	[0.758]	[0.554]	[0.735]	[0.852]	[0.454]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,772,321	3,745,254	3,745,254	3,772,321	3,745,254	3,745,254
Clusters (d)	214	214	214	214	214	214
Adj. R^2	0.737	0.835	0.923	0.737	0.835	0.923

Notes: C-prod-HS4 subsample: green products plus the non-green products in the 106 HS4 headings that contain at least one green code, so that each green product is compared with its four-digit neighbours. Full firm-level panel, excluding Hong Kong and Macao. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 35: CEM destinations — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0037)	−0.0014 (0.0041)	−0.0003 (0.0016)	−0.0006 (0.0011)	−0.0002 (0.0012)	−0.0004 (0.0003)
Asymptotic p	[0.543]	[0.734]	[0.837]	[0.576]	[0.857]	[0.201]
Bootstrap p	[0.610]	[0.824]	[0.887]	[0.609]	[0.878]	[0.241]
$EP_{dt} \times b_p$	−0.0041* (0.0024)	−0.0042 (0.0034)	−0.0002 (0.0013)	−0.0011 (0.0007)	−0.0006 (0.0010)	−0.0004 (0.0003)
Asymptotic p	[0.098]	[0.214]	[0.868]	[0.140]	[0.541]	[0.308]
Bootstrap p	[0.251]	[0.544]	[0.928]	[0.204]	[0.610]	[0.403]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	13,992,396	13,858,213	13,858,206	13,992,396	13,858,213	13,858,206
Clusters (d)	57	57	57	57	57	57
Adj. R^2	0.742	0.815	0.921	0.742	0.815	0.921
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0016 (0.0034)	−0.0016 (0.0035)	0.0000 (0.0019)	−0.0005 (0.0011)	−0.0003 (0.0011)	−0.0003 (0.0005)
Asymptotic p	[0.648]	[0.644]	[0.980]	[0.655]	[0.783]	[0.567]
Bootstrap p	[0.676]	[0.737]	[0.978]	[0.676]	[0.815]	[0.683]
$EP_{dt} \times b_p$	−0.0043* (0.0024)	−0.0055* (0.0030)	0.0001 (0.0013)	−0.0013* (0.0007)	−0.0011 (0.0009)	−0.0002 (0.0003)
Asymptotic p	[0.081]	[0.071]	[0.938]	[0.091]	[0.194]	[0.478]
Bootstrap p	[0.121]	[0.180]	[0.981]	[0.125]	[0.255]	[0.607]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	13,992,396	13,858,213	13,858,206	13,992,396	13,858,213	13,858,206
Clusters (d)	57	57	57	57	57	57
Adj. R^2	0.742	0.815	0.921	0.742	0.815	0.921

Notes: CEM-matched destinations subsample. Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). Quantity and unit-value regressions estimated on the subset with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 36: Deep vs. shallow agreements — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0022 (0.0035)	−0.0013 (0.0036)	−0.0004 (0.0011)	−0.0004 (0.0010)	−0.0001 (0.0010)	−0.0002 (0.0002)
Asymptotic p	[0.534]	[0.712]	[0.748]	[0.721]	[0.903]	[0.392]
Bootstrap p	[0.585]	[0.817]	[0.708]	[0.751]	[0.919]	[0.305]
$EP_{dt} \times b_p$	−0.0034 (0.0029)	−0.0034 (0.0034)	−0.0005 (0.0008)	−0.0006 (0.0010)	−0.0004 (0.0012)	−0.0003 (0.0003)
Asymptotic p	[0.246]	[0.330]	[0.572]	[0.549]	[0.712]	[0.389]
Bootstrap p	[0.499]	[0.601]	[0.771]	[0.739]	[0.785]	[0.379]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	5,262,293	5,201,289	5,201,282	5,262,293	5,201,289	5,201,282
Clusters (d)	23	23	23	23	23	23
Adj. R^2	0.751	0.844	0.932	0.751	0.844	0.932
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0016 (0.0032)	−0.0015 (0.0031)	−0.0002 (0.0018)	−0.0003 (0.0010)	−0.0002 (0.0009)	−0.0002 (0.0004)
Asymptotic p	[0.619]	[0.633]	[0.915]	[0.776]	[0.832]	[0.624]
Bootstrap p	[0.616]	[0.727]	[0.910]	[0.784]	[0.857]	[0.715]
$EP_{dt} \times b_p$	−0.0042 (0.0025)	−0.0043 (0.0028)	−0.0007 (0.0009)	−0.0009 (0.0008)	−0.0008 (0.0009)	−0.0003 (0.0003)
Asymptotic p	[0.113]	[0.141]	[0.431]	[0.253]	[0.401]	[0.294]
Bootstrap p	[0.173]	[0.294]	[0.755]	[0.386]	[0.492]	[0.336]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	5,260,451	5,199,451	5,199,444	5,260,451	5,199,451	5,199,444
Clusters (d)	23	23	23	23	23	23
Adj. R^2	0.751	0.844	0.932	0.751	0.844	0.932

Notes: Deep vs. shallow agreements subsample: restricted to destinations with at least one deep PTA in force. Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). Quantity and unit-value regressions estimated on the subset with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

R Minimum detectable effects

Table 37 reports the minimum detectable effect for the headline green and dirty coefficients under the empirical standard errors and the wild-cluster-bootstrap confidence intervals, expressed in both per-provision and per-standard-deviation terms. It is estimated on the collapsed panel, where the standard errors are larger than in the full panel, so the bounds it reports are looser than those quoted in Section 5.2.

The distinction it draws is worth stating explicitly. A null result can mean that the effect is absent or that the design cannot see it, and only an interval separates the two. On the green margin the bootstrap interval rules out effects larger than roughly three percent per provision at 95 percent confidence, and says nothing about anything smaller. For scale, Brandi et al. (2020) find that one additional liberalising environmental provision raises the green export share by about 17 percent (0.157 log points) in their aggregate bilateral panel of developing-country exporters. The comparison is only indicative: their unit is a liberalising provision, ours any environmental provision category, and their outcome a share rather than a within-firm log gap. On that reading, the full-panel upper bound of Section 5.2, 3.5 percent per provision, is roughly one quarter of their figure and the collapsed-panel bound in Table 37, 3.2 percent, roughly one fifth.

Table 37: How large an effect would the design need to detect it

Index	Margin	Asymptotic std. error	Minimum detectable effect (1 s.d.)	Bootstrap CI half-width (1 s.d.)	Bootstrap CI (per unit)
WB	Green	0.0070	4.64%	6.00%	[-1.85%, 3.18%]
WB	Dirty	0.0030	1.97%	2.40%	[-1.85%, 0.17%]
TREND	Green	0.0018	4.16%	3.60%	[-0.17%, 0.71%]
TREND	Dirty	0.0016	3.65%	3.56%	[-0.31%, 0.56%]

Notes: Collapsed panel, excl. HK/Macao. Standard deviation of the EP indices weighted by cell size over all 3,773,498 cells of the collapsed panel, never-treated destinations included (2.38 for WB, 8.16 for TREND). Minimum detectable effect = $2.8 \times$ asymptotic standard error, which understates the true MDE by the same factor by which the bootstrap widens the interval; the bootstrap half-width column is the relevant bound.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.